

Long-Term Investors, Demand Shifts, and Yields

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I exploit a Dutch reform in the regulatory discount curve that makes the liabilities of pension funds and insurance companies (P&Is) more sensitive to changes in 20-year interest rates but less so to longer maturity rates. Following the reform, P&Is reduced their longest maturity bond holdings but increased those with 20-year maturities, steepening the long end of the yield curve. Using the reform as a shock to identify price elasticities of demand at the sector level based on holdings across maturity buckets and time, I show that banks are more price elastic than other investors and absorb demand shocks. (*JEL* G11, G12, G18, G22, G23, G28)

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Recent literature shows that long-term investors, such as pension funds and insurance companies, affect yields (e.g., [Greenwood and Vayanos 2010](#); [Domanski, Shin, and Sushko 2017](#); [Greenwood and Vissing-Jorgensen 2018](#); [Klinger and Sundaresan 2019](#)). The findings in this literature are consistent

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with the preferred habitat view: demand for specific maturities creates price pressure on bond markets. Demand shocks by these investors can affect yields in case of inelastic demand (e.g., [Koijen and Yogo 2019](#)) that, for instance, arises in the presence of limits to arbitrage (e.g., [Vayanos and Vila 2021](#)).

Because of data limitations, the literature so far has primarily used only price data to study the implications of the preferred habitat view on yields. As a result, two important questions have been left largely unanswered. First, how important are demand shocks from long-term investors *quantitatively* in affecting bond yields? Second, what is the *mechanism* through which demand affects yields, and, more importantly, what does that tell us about the arbitrageurs in government bond markets?

To address these questions, I focus on a regulatory reform targeted at pension funds and insurance companies (henceforth: P&Is) in the Netherlands. As [Greenwood and Vissing-Jorgensen \(2018\)](#) point out, the reform affected long-term bond yields around the announcement day. My contribution is to study the underlying quantities to test and quantify the causal effect of demand shifts on government bond yields. More importantly, the unexpected nature of the reform allows for the identification of price elasticities of demand for several investor types. These demand elasticities give insight in the mechanism behind the observed yield effects because they show which investors respond more to changes in yields, thereby shedding light on the question of who are the arbitrageurs in government bond markets.

The Dutch P&I market is a useful laboratory to study the demand for long-dated assets and its effects on yields for three main reasons. First, the occurrence of a regulatory reform facilitates the clean identification of demand shifts and their effect on yields. The second reason is the availability of detailed data on bond and derivative positions at the institutional level on the one hand, and data about the investors' liabilities on the other hand. While data on bond holdings are widely available for various investor types, detailed data on derivative positions and liabilities are typically scarce.¹ Yet, P&Is hedge a substantial amount of their interest rate risk with derivatives. To accurately estimate their demand for long-dated assets, one must have data on both bond and derivative positions. To explain the drivers of demand for long-dated assets, data on the liability structure of P&Is is also key. Third, the Dutch P&I market is economically important because the pension sector alone already covers 58% of the total assets of pension funds in the euro area ([OECD 2020](#)).²

I exploit a change in the regulatory discount curve to value liabilities that the Dutch regulator announced and directly implemented on July 2, 2012.

¹ Detailed data on bond holdings, derivative positions, and liabilities exist for the U.S. insurance sector and are available for research (see, e.g., [Sen 2023](#)) but these data do not (publicly) exist for U.S. pension funds.

² In 2019, the assets under management (AUM) of pension funds equaled approximately €1.75 trillion, or 192%, of gross domestic product (GDP) and represented 58% of the total assets of pension funds in the euro area ([OECD 2020](#)). The AUM of insurance companies equaled €0.51 trillion and represented 6% of the total assets of insurance companies in the euro area ([ECB 2021](#)).

This reform altered the way the discount curve is calculated: it now relies *more* on the 20-year interest rate and *less* on rates for longer maturities. For maturities up to 20 years, the curve uses market interest rates. For those over 20 years, it uses a weighted average of the 20-year rate and a fixed rate called the ultimate forward rate (UFR), which is higher than market interest rates. As such, the reform lowered the calculated value of the liabilities. At the same time, liabilities became more sensitive to changes in the 20-year rate, but less sensitive to changes in longer maturity rates.

Why does a change in the regulatory discount curve affect demand for long-dated assets? This demand arises for two reasons: economic and regulatory hedging incentives. The long-term nature of liabilities creates a natural preference for long-term bonds from a liability hedging perspective (Sharpe and Tint 1990; Campbell and Viceira 2002). Regulatory hedging incentives are important when the regulatory framework diverges from the economic state (e.g., Koijen and Yogo 2015, 2016; Sen 2023), because the regulator uses the regulatory discount curve to assess the solvency positions of P&Is. In turn, solvency positions determine the amount pension funds can pay out to retirees or insurance companies to shareholders in terms of dividends. As a result, incentives to hedge the regulatory discount curve rise if the curve diverges from the economic discount curve.

I report the following three main findings. First, consistent with the predictions of a mean-variance framework in an asset-liability context with regulatory constraints, I observe that P&Is with substantial exposure to the regulatory reform, that is, those with long-term liabilities, significantly reduced their bond holdings with a remaining time-to-maturity of at least 30 years (henceforth for brevity: 30-year bond holdings). The aggregate decrease in 30-year bond holdings is economically large: the total decline in demand equals 30% relative to the pre-reform 30-year bond holdings. At the same time, P&Is increase their aggregate bond holdings with maturities close to 20 years by 19% relative to the pre-reform holdings. Both effects are stronger for P&Is that are closer to their solvency constraints, because they have a stronger incentive to hedge the regulatory rather than the economic discount curve. Additionally, I find a decline in the implied duration of swap portfolios after the reform and I show large exposures to the 20-year swap rate but small exposures to longer maturities based on detailed data on swap positions that became available as part of the European Market Infrastructure Regulation (EMIR) in 2017. Finally, the direct capital relief from the reduction in the liability values led to an increase in their allocation to equities and high-yield government bonds. Overall, the results show that the regulatory reform led to a reduction in economic hedges of interest rate risk and to a rise in risky asset exposures at the same time.³

³ I perform several robustness checks to show that these findings cannot be explained by the ECB's asset purchasing programs (APP) or the chosen length of the pre and post periods.

Second, I show that the decreased demand for 30-year bonds and increased demand for 20-year bonds led to a steepening of the long-end of the Dutch yield curve. Because P&Is hold a large share of the Dutch government debt outstanding, I am able to identify the mechanism behind these yield curve effects. To achieve this goal, I estimate demand curves as in [Kojien and Yogo \(2019\)](#) for the other institutions that hold Dutch debt. I use the regulatory reform as an instrument for bond yields, because the regulatory reform created an exogenous demand shift by the Dutch P&I sector that only affected the demand of other investor types through its effect on prices. I justify the plausibly exogenous nature of the reform by the fact that regulator announced and directly implemented the reform 3.5 years before it became part of the new Solvency II regulatory framework in 2016. Even though the discussions at the time evolved around the convergence of regulatory discount rates to a stable level and P&Is may have anticipated the regulatory reform as a result, they did not know the implementation date and the determinants of the shape of the UFR, such as its level and slope. As such, how the reform would affect different maturities is unclear *ex ante* and in particular the *ex post* reliance of the curve on the 20-year market interest rate. The less anticipated nature of the reform becomes evident by looking at the sudden increase in the 30-20 year yield spread on the announcement and implementation date of the UFR in [Figure 1](#). The unexpected nature of the reform to the market is further motivated by the regulator prohibiting the P&I sector to trade on information discussed

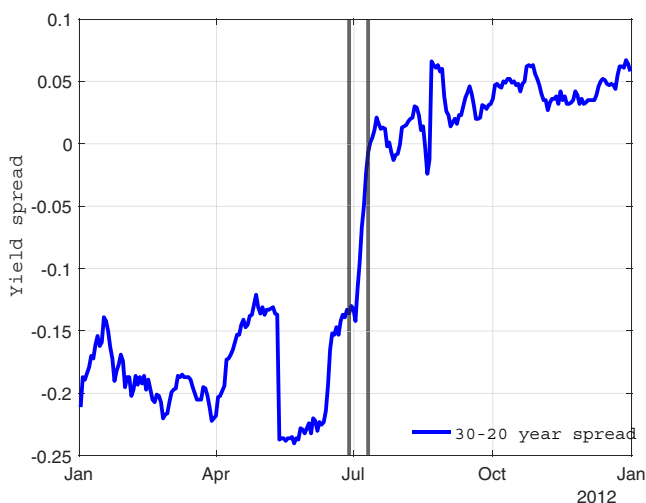


Figure 1

Dutch 30-20 year yield spread: UFR

This graph shows the Dutch 30-20 year government bond yield spread in percentage points around the announcement (and implementation) date of the UFR. The vertical lines represent 5 days before and after the announcement (and implementation) of the UFR on July 2, 2012.

during meetings in the weeks leading up to the reform.⁴ That said, to the extent that P&Is anticipated a lower sensitivity of their liabilities to very long-term interest rates, the estimated changes in yields may understate the true effects and demand elasticities provide an upper bound of the true elasticities.

The demand elasticities that I estimate are at the sector level, using data across maturities and across time. I find that banks are most price elastic and largely responsible for absorbing the demand shock. Indeed, after the reform, banks increased their 30-year bond holdings by 18% of the total amount outstanding (TAO) while at the same time they reduced their holdings toward 20-year ones by 9% of the TAO. To better understand why banks absorb demand shocks, I show that P&Is predominantly trade swaps with banks as the counterparty. Moreover, banks hold the opposite exposures compared to P&Is in the swap market. These findings suggest that banks' detailed information about demand for long-dated assets by the P&I sector allows them to quickly react to uninformed demand shocks from this sector.

In response to the reform, P&Is rebalanced not only across Dutch government bonds but also across other safe European bonds, primarily those of France and Germany. As a result, the reform also led to a steepening of the long-end of the yield curve in these countries, although the magnitude of this effect was only half of that observed in the Netherlands. The fact that the reform had a much stronger impact on Dutch government bond markets suggests that the Dutch market is segmented from other safe European bond markets, as otherwise, the yields in various safe European bond markets would move in lockstep in response to the reform. Consistent with this observation, I find similar price elasticities of demand whether I explicitly allow investors to operate only in the Dutch government bond market or when I allow them to substitute between Dutch and other safe European bond markets.

Finally, to generalize my findings, I provide suggestive evidence for the effects of the regulatory reform on yields at the European level. The UFR is an important aspect of the EU Solvency II regulation for insurers in Europe that was announced in August 2015 and took effect on January 1, 2016. Relying on the home-bias found in euro area investors' portfolios (e.g., [Kojen et al. 2017, 2021](#)), I investigate the effects of the UFR on a broad panel of 20 European countries and define the demand for long-term bonds by the size of the insurance sector relative to the total debt outstanding of a country. My estimates suggest that for countries with a small insurance sector changes in yields were negligible; while those with large insurance sectors experienced increases in the 30-20 year yield spread that are of similar magnitude to the Netherlands. Using holdings data at the insurer-country level in the euro area, I confirm that insurance companies substantially increased their exposure to

⁴ One specific insurer, Delta Lloyd, did not comply and as a result, received a €22.7 million fine. For details pertaining to *De Nederlandsche Bank v. Delta Lloyd Levensverzekering NV (DL Leven)*, see [Rotterdam District Court \(2015\)](#) ruling.

20-year bonds after Solvency II, consistent with the observed increase in the 30-20 year yield spreads.

My findings have important policy implications. First, they show that the regulatory framework of investors has direct consequences for the governments' costs of borrowing. Second, they show that the regulation of long-term investors spills over to the banking sector. As pointed out by [Kojien and Yogo \(2022\)](#), regulation of one sector is typically studied in isolation from other sectors by regulators. My findings, however, suggest that regulators should take the regulatory framework of banks into account when analyzing the introduction of new regulatory policies targeted at long-term investors.

This study contributes to the preferred habitat theory ([Culbertson 1957](#); [Modigliani and Sutch 1966](#); [Vayanos and Vila 2021](#)) that posits that some investors prefer specific maturities and that their demand for bonds with those maturities influences the interest rates at those maturities. [Greenwood and Vayanos \(2014\)](#) find that supply shocks affect yields because they change the duration risk of arbitrageurs. Similarly, [Guibaud, Nosbusch, and Vayanos \(2013\)](#) show that an increase in investors with a longer investment horizon renders long-term bonds more expensive and increases their optimal supply by the government. I contribute to this literature by providing direct evidence in favor of the preferred habitat theory.

My findings also contribute to the work that empirically tests the implications of the preferred habitat or the demand-based view. For instance, [Domanski, Shin, and Sushko \(2017\)](#) argue that the “hunt-for-duration” of insurance companies might have amplified the decline in bond yields in the euro area. Similarly, [Klinger and Sundaresan \(2019\)](#) explain that the negative 30-year U.S. swap spreads are the result of underfunded pension plans optimally using swaps for duration hedging. Additionally, [Greenwood and Vayanos \(2010\)](#) show that changes in regulation which increase the incentive to buy (sell) long-term bonds result in a decline (increase) in long-term yields around policy announcement days.⁵ Closest to my paper, [Greenwood and Vissing-Jorgensen \(2018\)](#) also use the regulatory reform that is studied here to investigate the impact of P&I demand on bond yields. The key contribution to their work is that I bring in quantities which allow me to explicitly test and quantify the causal impact of hedging demand on government bond yields. I further complement their work by using the reform to back out the aggregate and sector-specific price elasticities of demand, shedding light on the market segmentation in government bond markets.

My study also contributes to the growing demand-based asset pricing literature ([Kojien and Yogo 2019](#)). [Kojien et al. \(2017, 2021\)](#) apply this model to study the effects of quantitative easing (QE) on yields in the euro area and [Bretscher et al. \(2024\)](#) study the implications of inelastic demand for corporate

⁵ A large body of literature provides evidence of the demand-based view for stocks (e.g., [Shleifer 1986](#); [Wurgler and Zhuravskaya 2002](#); [Greenwood 2005](#); [Chang, Hong, and Liskovich 2015](#); [Pavlova and Sikorskaya 2023](#)).

bond pricing. A key challenge in demand-based asset pricing is to find clean instruments for prices. I make progress in this direction by using an exogenous demand shift as an instrument to establish a causal link between demand and asset prices. Compared to the traditional instrument in [Kojien and Yogo \(2019\)](#) that is based on unobservable investment mandates, I use an instrument that is empirically observable. Furthermore, I employ an instrument that is largely free from anticipation effects, unlike for instance, the capital key instrument used in [Kojien et al. \(2021\)](#) that was known in advance of the QE programs.

While recent literature in demand-based asset pricing directly incorporates detailed portfolio holdings to understand asset prices, earlier works have recognized the price impact of (institutional) investor demand. For instance, [Wermers \(1999\)](#), [Nofsinger and Sias \(1999\)](#), [Gompers and Metrick \(2001\)](#), and [Cohen, Gompers, and Vuolteenaho \(2002\)](#), among others, demonstrate that stock prices rise (fall) with large institutional purchases (sales). In currency markets, studies by [Cai et al. \(2001\)](#), [Evans and Lyons \(2002\)](#), and [Froot and Ramadorai \(2005\)](#), among others, have linked large investor flows to transitory exchange rate dynamics. Likewise, in the international real estate market, research by [Badarinza and Ramadorai \(2018\)](#), [Gorback and Keys \(2020\)](#), and [Favilukis and Van Nieuwerburgh \(2021\)](#) document that foreign demand influences real estate prices. While these studies highlight price impact of investor demand, the underlying mechanisms remain largely unexplored. My paper addresses this gap by considering the demand from all investors simultaneously to understand the behavior of investors that ultimately have to absorb these large demand shocks. My framework therefore provides a useful tool for understanding cross-sectional variation in price impact across markets and asset classes.

Additionally, my study links to the intermediary asset pricing literature, which directly models intermediaries and how they matter for asset prices (e.g., [He and Krishnamurthy 2013](#); [Adrian, Etula, and Muir 2014](#); [Haddad and Muir 2021](#); [Baron and Muir 2022](#)). The focus in these studies is on the link between asset prices and constraints of (investment) banks, hedge funds, and broker dealers. Instead, I provide evidence for the importance of regulatory constraints for the prime holders of government debt: the P&I sector.

Finally, my findings link to the literature on the effects of regulation for the investment and hedging behavior of long-term investors. [Ellul, Jotikasthira, and Lundblad \(2011\)](#) show that insurance companies' regulatory constraints induce fire sales; [Becker and Ivashina \(2015\)](#) find that insurers are biased toward high-yielding bonds conditional on credit ratings; [Becker, Opp, and Saidi \(2022\)](#) show that insurance companies sold fewer poorly rated MBS after elimination of capital requirements on MBS; and [Ellul et al. \(2022\)](#) find that variable annuities induce more risky and illiquid bond holdings. [Andonov, Bauer, and Cremers \(2017\)](#) show that U.S. public pension plans invest riskier to report a better funding status. Closest to my paper is [Sen \(2023\)](#), who finds that insurance companies only hedge interest rate risk and equity risk when

the regulatory framework recognizes such risks. My findings also show that regulatory incentives are key in shaping the hedging decisions of P&Is.

1. Institutional Setting: Ultimate Forward Rate

In this section, I describe the regulatory framework of both pension funds and insurance companies and explain the regulatory reform in detail.

1.1 Regulatory framework P&Is

Pension funds and insurance companies are regulated by the Dutch Central Bank (DNB) and in this section I briefly describe both regulatory frameworks. For both sectors, it is important to note that the valuation of assets has been mark-to-market since 2007.⁶

A pension fund's solvency position is assessed by computing its funding ratio, or the assets divided by the liabilities. A pension fund's required funding ratio is based on a pension fund's risk profile and is calculated such that the probability that capital (the funding ratio) falls below zero (100%) on a 1-year horizon equals 2.5%. In case a pension fund is not compliant with the required funding ratio, it files a recovery plan to the supervisor. Additionally, pension funds are not allowed to index pension rights if the funding ratio falls below a minimum threshold of 104.2%. If the funding ratio falls below 90% a reduction in accrued benefits may be required. Note that the regulatory framework of Dutch pension funds is very different from its U.S. counterpart, where risk-based capital requirements are absent (Boon, Brière, and Rigot 2018).

An insurance company's solvency position is assessed by computing its solvency ratio, or the available capital divided by the required capital. Prior to Solvency II, the solvency ratios of insurance companies were not risk-based and the DNB required capital to equal 4% of the value of the liabilities. At the introduction of Solvency II, it required capital to be computed like for pension funds, except that it calculated required capital such that the probability of capital falling below zero on a 1-year horizon equaled 0.5%, rather than the 2.5% for pension funds. In case an insurance company is not compliant with the required solvency requirements, it also files a recovery plan to the supervisor. Dividend policies are typically based on internal target solvency ratios. For instance, Allianz only pays out dividends if the solvency ratio exceeds 160%.⁷

1.2 Regulatory discount curve

The regulator uses the regulatory discount curve to estimate the liability value and hence the solvency positions of P&Is. Prior to the start of July 2012,

⁶ A mark-to-market valuation of the assets is in stark contrast with the life insurance industry in the United States, where historical cost accounting is still commonly used across states (Ellul et al. 2015).

⁷ See https://www.allianz.com/en/investor_relations/share/dividend.html.

the DNB based the regulatory discount curve on the euro swap curve for all maturities and used market interest rates up to a maturity of 50 years. Afterward, it extrapolated the interest rates for maturities beyond 50 years from the last observed forward rate. Because it based the regulatory discount curve on market interest rates only, the curve equaled the economic discount curve. The DNB announced and directly implemented a change in the regulatory discount curve on July 2, 2012 in anticipation of the EU introducing its Solvency II regulatory framework for insurance companies in 2016. The new curve is similar to the regulatory discount curve in Solvency II that would become applicable to all European insurers. The DNB announced a similar regulatory discount curve for pension funds on September 24, 2012.⁸

The DNB's new regulatory discount curve uses an extrapolation method based on the ultimate forward rate (UFR) that is the convergence of long interest rates to a stable level. In essence, this new curve uses market interest rates up to a maturity of 20 years, and the DNB determines interest rates with maturities longer than 20 years by using a weighted average between the market interest rates and a fixed rate, the UFR. The main argument to justify the implementation of the UFR was that the market for long durations is less liquid and only a few securities with such long durations existed. As a result, the DNB regarded the implied market interest rates as unreliable: a discount curve purely based on market data was highly sensitive to liquidity shocks and therefore also the solvency positions of P&Is. A regulatory discount curve based on the UFR solved this issue by making the long-end of the curve less dependent on market interest rates.⁹

Formally, the DNB constructed the regulatory discount curve as follows:

1. The euro swap rates for maturities of 1 to 10, 12, 15, and 20 years are converted to zero-coupon interest rates by means of bootstrapping. For nonobservable swap rates, the DNB estimates zero interest rates by assuming constant forward rates.
2. Forward rates exceeding 20 year maturities are a weighted average between the 20-year forward rate and the UFR. The weight increases linearly in maturity, and the level of the UFR equals 4.2%. At a maturity of 60 years, forward rates equal the UFR:¹⁰

⁸ Despite the UFR for pension funds being announced only 85 days later, widespread public expectations at the time hoped that the UFR would be implemented for pension funds as well. This expectation was clearly reflected when Dutch House of Representatives members proposed a parliamentary motion on July 5, 2012, just 3 days after the announcement for insurers, to introduce the UFR for pension funds (*Tweede Kamer der Staten-Generaal 2012*). This anticipatory effect also explains the lack of a noticeable yield response on September 24, 2012 (Figure 1).

⁹ Further details about the UFR are in [Internet Appendix IA3.2](#)

¹⁰ For pension funds, the regulatory discount curve is slightly different and uses the corresponding market forward rate $f_{t,h-1}^h$ instead of the 20-year forward rate for maturities of 25, 30, 40, and 50 years.

$$f_{t,h-1}^{h,*} \begin{cases} f_{t,h-1}^h & \text{if } 1 \leq h \leq 20, \\ (1 - w^{UFR}(h)) \times f_{t,19}^{20} + w^{UFR}(h) \times UFR & \text{if } 20 < h < 60, \\ UFR & \text{if } h \geq 60. \end{cases} \quad (1)$$

3. The DNB computes the zero-coupon interest rates $y_t^{(h)}$ as follows:

$$(1 + y_t^{(h)})^h = \prod_{n=1}^h (1 + f_{t,n-1}^{n,*}) \quad \text{for } h = 1, 2, \dots, 120. \quad (2)$$

The regulatory discount curve with the UFR has three important effects. First, the UFR decreases current liability values as liabilities are discounted against higher rates. Second, the UFR makes the regulatory discount curve less sensitive to parallel shifts in interest rate changes for maturities beyond 20 years. Panel A of Figure 2 displays these two effects. Third, the UFR results in a higher exposure to changes in the 20-year interest rate as shown in panel B of Figure 2. A localized negative shock in the 20-year market interest rate as reflected by the blue dashed line reduces all the regulatory discount rates for maturities beyond 20 years that makes the liabilities particularly sensitive to changes in the 20-year interest rate under the new regulatory framework.¹¹ Internet Appendix IA3.1 shows that in practice these changes had economically important effects on regulatory liability values.

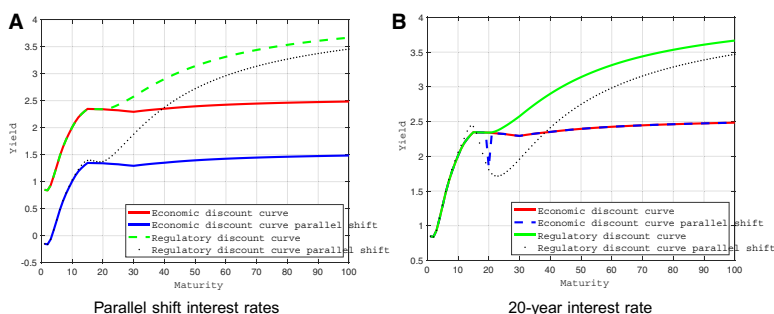


Figure 2

Change in the regulatory discount curve

Panel A shows the economic discount curve (solid red line) and the regulatory discount curve (dashed green line) at implementation of the UFR on July 2, 2012. The graph also shows the economic (solid blue line) and regulatory (dotted black line) discount curve after a parallel shock in market interest rates of $\Delta y_t = -1\%$. Panel B shows the economic discount curve (solid red line) and the regulatory discount curve (solid green line) at implementation of the UFR on July 2, 2012. The graph also shows the economic (dashed blue line) and regulatory (dotted black line) discount curve after a change in the 20-year market interest rate of $\Delta y_t^{(20)} = -0.5\%$.

¹¹ Note that a shock that affects the 20-year interest rate in isolation is not common and is only used for illustrative purposes.

2. Model Intuition

To better micro-found how P&Is changed their asset demand following the reform, I derive my main testable predictions from a simple partial equilibrium mean-variance optimization framework with assets and liabilities. The key driver of my predictions is the assumption that P&Is care about both their economic and regulatory solvency positions. The full version of the model is in [Internet Appendix IA4](#), but I give a summary here to get the main intuition across.

For the assets, I assume that P&Is can invest in a risky asset S with corresponding return r_{t+1}^S , and in a set of safe European bonds $B(h)$ with corresponding maturity h and return $r_{t+1}^{(h-1)}$. The vector of bond returns is denoted by $\mathbf{r}_{t+1}^B$. I assume that the yield curve can be determined using this set of bonds. The risk-free rate is denoted by r_f and total assets by A_t . The assets therefore evolve as $A_{t+1} = \left(1 + r_f + w_t^S(r_{t+1}^S - r_f) + \mathbf{w}_t^{B'}(\mathbf{r}_{t+1}^B - r_f \mathbf{1})\right) A_t$.

Throughout, the simplifying assumption is that P&Is consider all safe European bonds as close substitutes and therefore the only relevant dimension is maturity. This is a simplification because P&Is hold safe European bonds from various countries even though their yields diverge. If P&Is consider these bonds to be perfect substitutes, then P&Is would only hold (buy) those safe European bonds with the highest yields (lowest prices). Even though this is a simplifying assumption, the object of study is the *change* in holdings following the reform. The underlying assumption, therefore, is that P&Is do not discriminate across countries in response to the reform. This is a plausible assumption because within the regulatory framework, safe European government bonds are treated in the same way. That is, hedging one unit with a bond from one safe European country versus another has the same impact on the regulatory solvency position. In the next section, I confirm that P&Is rebalanced in a similar way within various safe European bond markets in response to the reform.

For the liabilities, I assume that P&Is have to pay out a fixed and promised set of cash flows for each maturity h , $CF(h)$.¹² The return on the liabilities is denoted by $\mathbf{r}_{t+1}^L$. Because of the promised nature of the cash flows, the economic value of the liabilities is determined using market interest rates, and hence $\mathbf{r}_{t+1}^L = \mathbf{r}_{t+1}^B$. The economic value of the liabilities therefore evolves as $L_{t+1}^E = \mathbf{a}_t'(\mathbf{1} + \mathbf{r}_{t+1}^B)L_t^E$, where $\mathbf{a}_t$ is a vector that contains the cash flow distribution of the liabilities. Intuitively, $a_{it}(h)$ is the fraction of the total liabilities that have to be paid in h years from now.

The regulatory value of the liabilities diverges from its economic counterpart in that its sensitivity to market interest rates is different. The regulatory value of the liabilities evolves as $L_{t+1}^R = \mathbf{a}_t'(\mathbf{1} + \boldsymbol{\xi}_L' \mathbf{r}_{t+1}^B)L_t^R$, where the sensitivity of the

¹² This assumption is realistic, because the pension funds are defined benefit in nature, and insurance companies were not allowed to use variable annuities until 2016.

regulatory discount curve to market interest rates is defined by ξ_L and has the same length as the set of bonds. This sensitivity means that the economic and regulatory value of the liabilities are identical if $\zeta_L(h)=1$ for all maturities h , which was the case prior to the reform. On the other hand, the regulatory value of the liabilities is insensitive to changes in the interest rate if $\zeta_L(h)=0$ for all maturities h . An example is a regulatory discount curve that is based on a fixed rate for all maturities. The reform implied that $\zeta_L(h)$ became larger than one for maturities close to 20 years, equivalent to an increased exposure of the regulatory liabilities to the 20-year market interest rate. At the same time, $\zeta_L(h)$ converges to zero for 30-year maturities, or a reduced exposure of the regulatory liabilities to the 30-year market interest rate.

I further assume that P&Is have mean-variance preferences over the assets minus liabilities, which is similar to Sharpe and Tint (1990) and Hoevenaars et al. (2008), and creates an incentive to reduce the volatility in the economic mismatch between assets and liabilities, or the volatility in the economic funding ratio. Following Koijen and Yogo (2015, 2016); Sen (2023), I also assume that P&Is care about the volatility in the regulatory mismatch between assets and liabilities, or their regulatory funding ratio. The reason is that P&Is have to make important decisions based on the regulatory funding ratio, such as the amount of dividends to pay to shareholders or the ability to index pension rights. The economic funding ratio is denoted by $F_t^E = A_t/L_t^E$ and the regulatory funding ratio by $F_t^R = A_t/L_t^R$. A simplified version of the optimization problem of assets over liabilities then equals:

$$\max_{w_t} \mathbb{E}_t \left[\frac{A_{t+1}}{A_t} \right] - \frac{\gamma}{2} \text{Var}_t \left[\frac{A_{t+1}}{A_t} - \frac{L_{t+1}^E}{A_t} \right] - \frac{\lambda(F_t^R)}{2} \text{Var}_t \left[\frac{A_{t+1}}{A_t} - \frac{L_{t+1}^R}{A_t} \right], \quad (3)$$

where γ equals the risk-aversion parameter, and $\lambda'(F_t^R) < 0$; or in other words, P&Is care more about the regulatory funding ratio when its low as opposed to when its high.¹³

Solving for the optimal portfolio weights results in:

$$w_t^{S*} = \frac{\mathbb{E}_t[r_{t+1}^S - r_f]}{(\gamma + \lambda(F_t^R)) \text{Var}_t[r_{t+1}^S]}, \quad (4)$$

speculative portfolio

$$\begin{aligned} w_t^{B*} = & \underbrace{\frac{\mathbb{E}_t[\mathbf{r}_{t+1}^B - r_f \mathbf{1}]}{(\gamma + \lambda(F_t^R)) \text{Var}_t[\mathbf{r}_{t+1}^B]}}_{\text{speculative portfolio}} + \underbrace{\frac{\gamma}{\gamma + \lambda(F_t^R)} \frac{1}{F_t^E} \mathbf{a}_t \circ \frac{\text{Cov}_t[\mathbf{r}_{t+1}^B, \mathbf{r}_{t+1}^L]}{\text{Var}_t[\mathbf{r}_{t+1}^B]}}_{\text{economic hedging portfolio}} \\ & + \underbrace{\frac{\lambda(F_t^R)}{\gamma + \lambda(F_t^R)} \frac{1}{F_t^R} (\mathbf{a}_t \circ \xi_L \circ \frac{\text{Cov}_t[\mathbf{r}_{t+1}^B, \mathbf{r}_{t+1}^L]}{\text{Var}_t[\mathbf{r}_{t+1}^B]})}_{\text{regulatory hedging portfolio}}. \end{aligned} \quad (5)$$

¹³ To keep the model tractable, the functional form of $\lambda(F_t^R)$ is a reduced form of the strict constraint that the funding ratio should be higher than a certain threshold (e.g., Leibowitz and Henriksson 1989).

The optimal demand for the risky asset consists of speculative demand only, because the liabilities are valued using the yield curve, and the safe bonds are assumed to be independent of the risky asset. The bond portfolio consists of three components: the speculative demand, the economic hedging demand, and the regulatory hedging demand. The economic (regulatory) hedging demand equals the desired hedge against changes in the economic (regulatory) liability value. The heterogeneity in demand for long-term bonds across P&Is depends on two main factors. First, the demand for long-term bonds depends on the distribution of the cash flow payments: if many of the liabilities are due in 30 years, that is, a high $a_t(30)$, the hedging demand for 30-year bonds is higher, all else equal. Second, the demand for long-term bonds depends on the weight assigned to the economic versus regulatory hedging demand. This weight depends on the relative magnitudes of $\lambda(F_t^R)$ and γ that is driven by solvency positions F_t^R : the closer P&Is are to the regulatory constraint, the stronger the desire to hedge the regulatory funding ratio and hence a higher weight is assigned to the regulatory hedging portfolio. For instance, if a large weight is assigned to the regulatory hedging portfolio (large $\lambda(F_t^R)$) and the regulatory discount curve is insensitive to changes in 30-year market interest rates, that is, $\zeta(30)=0$, the total demand for 30-year bonds approaches zero.

In [Internet Appendix IA4](#), I formally derive the model implications of the regulatory reform by taking the difference between the model-implied optimal demand after and before the reform. This results in the following model implications:

Prediction 1 - *P&Is with long liability durations reduce their 30-year bond holdings and increase those with 20-year maturities more compared to P&Is with short liability durations.*

Intuitively, the first prediction simply states that P&Is with long liability durations, so those that pay out in the far future, are more affected by the reform and hence have a stronger incentive to move away from 30-year bond holdings toward 20-year ones.

Prediction 2 - *P&Is with long liability durations increase their risky asset holdings more compared to P&Is with short liability durations.*

P&Is with long liability durations saw a sharper decline in their liability values and hence experienced the largest capital relief. Because $\lambda(\cdot)$ is a convex and decreasing function of the regulatory funding ratio, those P&Is have the strongest incentive to increase their speculative portfolio.

Prediction 3 - *P&Is close to their solvency constraint reduce their 30-year bond holdings and increase those with 20-year maturities more compared to unconstrained P&Is.*

The more constrained investors are, the stronger the incentive to hedge the regulatory discount curve which implies moving away from 30-year bonds to those with 20-year maturities.

In [Internet Appendix IA4](#), I close the model by introducing an arbitrageur and impose market clearing to study the model-implied effects on the yield curve ([Vayanos and Vila 2021](#)). I then calibrate the model and show that the model-implied yield curve steepened because of the reform in [Figure IA3](#). The steepening of the model-implied yield curve corresponds to the observed change in the yield curve around the implementation date of the reform as shown in [Figure IA4](#).

3. Data

In this section, I describe the data sources that I use for my analysis ([Subsection 3.1](#)) and provide summary statistics for the sample ([Subsection 3.2](#)).

3.1 Constructing the data set

I use data on Securities Holdings Statistics (SHS) for insurance companies and pension funds in the Netherlands over the period 2009q1-2019q4. All institutions that report are domiciled in the Netherlands and the regulator decides which institutions have to report in order to have sufficient coverage in terms of AUM for every sector. Institutions have to report their holdings of all securities, both foreign and domestic, to the regulator on a quarterly basis.¹⁴ DNB gathers holdings data to compute, among other things, the Dutch balance of payments, international investment positions, and the financial accounts.¹⁵

The data provide bond and stock holdings with International Securities Identification Numbers (ISIN). The holdings of stocks are available in the number of shares and market values. The holdings of bonds are available in both nominal and market values. The SHS database is then linked to the Centralised Securities Database (CSDB). The aim of this database is to hold accurate information on all individual securities relevant for the statistical purposes of the European System of Central Banks, [ECB \(2010\)](#). I obtain the following relevant market information from the CSDB database: debt type, maturity dates, coupon rates, coupon frequencies, coupon type (e.g., fixed, floating or zero-coupon), last coupon payment date, yield-to-maturity (YTM), prices, and total amount outstanding. I compute bond durations using YTM and measure credit risk for corporate bonds using the distance to default (DTD) available from the Credit Research Initiative at the National University of Singapore. Credit risk for government bonds is measured using credit ratings from Fitch.

Next, I link the holdings data to the supervision databases. The supervision databases are from mandatory annual and quarterly statements that P&Is report

¹⁴ All institutions report their foreign holdings on a monthly basis, but some report domestic holdings on a quarterly basis. Because Dutch P&Is hold significant domestic fixed income securities, I use quarterly data to ensure consistency. For details on reporting requirements, see De Nederlandsche Bank website <https://statistiek.dnb.nl/statistiek/index.aspx>.

¹⁵ [Timmer \(2018\)](#) also uses SHS data at the institutional level for Germany.

to the DNB. P&Is have to report, among other things, solvency positions, liability durations, and asset allocations as well as the value of the assets and liabilities. For pension funds, I take the reported funding ratios directly from the files. For insurance companies, I convert the solvency ratios to funding ratios, because the model makes predictions based on funding ratios.¹⁶ Formally, prior to Solvency II, solvency ratios equaled $SR = \frac{A-L}{0.04L}$ which meant that the funding ratio equaled $\frac{A}{L} = 0.04 * SR + 1$. The solvency ratios under Solvency II are more complex, and hence I hand-collect data on the assets and the liabilities for each insurer to compute the funding ratios manually.

3.2 The sample

The total sample covers 42 pension funds, 12 life insurers, and 27 nonlife insurers. This group of institutional investors represents around 80%-90% of the AUM for all institutional investors domiciled in the Netherlands.

I only analyze investors' direct holdings, that is, investments that are not made via other investor types such as mutual funds. The data, unfortunately, do not allow for a linkage between the indirect holdings of investors to their direct holdings, except for the two largest pension funds and the two largest insurance companies. For these P&Is, I know their shares in mutual funds so I can use the reported holdings of mutual funds to obtain their indirect holdings. Table 1 shows that after correcting the holdings for these four P&Is, the fraction of assets that are incorporated in my analysis equals on average 85% of the total AUM (direct+indirect).

Table 1 also shows that life insurance companies are the largest in terms of average AUM, followed by pension funds.¹⁷ The average allocation to government bonds is 49%, 45%, and 38% for life insurers, nonlife insurers, and pension funds, respectively. Insurers have an allocation to corporate bonds of 34% and pension funds of 21%. Life insurers and pension funds also have the longest duration of their fixed income portfolios. The average bond durations are 8.4 and 7.3 years for life insurers and pension funds, respectively, while the duration equals 4.64 for nonlife insurers. Pension funds have the largest equity allocation at 34%, while insurers invest on average 12% of their assets in equities.

Moving to the liability side of P&Is, life insurers and pension funds have the longest liability durations at 11.8 and 17.9 years, respectively. The liability duration of nonlife insurers is much shorter at 4.2 years. The average inverse of the funding ratio of pension funds equals 92.5% and 91.5% for insurers, which is equivalent to a funding ratio of 108% and 109%, respectively.¹⁸

¹⁶ Notice that solvency ratios can be converted into funding ratios and vice versa.

¹⁷ See the [Internet Appendix \(Table IA1\)](#) for the extended summary statistics table.

¹⁸ Notice that pension funds in the Netherlands are much better funded compared to pension funds in the United States, potentially explained by the fact that Dutch pension funds are subject to risk-based capital requirements, whereas U.S. pension funds are not (Boon, Brière, and Rigot 2018).

Table 1
Summary statistics

A. AUM							
All assets	Mean	SD	p50	Direct assets	Mean	SD	p50
Life insurers	25,486	20,398	25,903	Life insurers	18,822	15,771	18,208
Nonlife insurers	1,507	1,380	957	Nonlife insurers	1,322	1,359	741
Pension funds	17,578	43,918	6,244	Pension funds	14,761	40,265	3,964
B. Asset allocation							
Government bonds	Mean	SD	p50	Corporate bonds	Mean	SD	p50
Life insurers	49.41	12.53	50.00	Life insurers	33.00	11.70	30.67
Nonlife insurers	45.32	19.75	44.84	Nonlife insurers	34.44	22.95	34.94
Pension funds	37.73	17.20	34.58	Pension funds	20.93	13.33	19.75
Stocks	Mean	SD	p50	Bond duration	Mean	SD	p50
Life insurers	10.08	9.92	6.71	Life insurers	8.72	2.61	8.74
Nonlife insurers	13.75	20.26	3.77	Nonlife insurers	4.47	1.66	4.59
Pension funds	34.13	16.86	36.49	Pension funds	7.29	2.22	7.02
C. Liability information							
Liability duration	Mean	SD	p50	Inverse funding ratio	Mean	SD	p50
Life insurers	11.77	3.95	12.56	Life insurers	91.60	3.42	92.26
Nonlife insurers	4.20	2.62	3.47	Nonlife insurers	91.37	4.01	92.48
Pension funds	17.94	2.96	17.70	Pension funds	92.53	9.43	92.59

This table shows summary statistics on the AUM (panel A), the asset allocation (panel B), and the liability information (panel C) of the P&I sector. In particular, I report the AUM of all assets, AUM of directly reported assets, allocation to government bonds, allocation to corporate bonds, allocation to stocks, bond duration, liability duration, and the solvency positions measured by the funding ratio. The asset allocation and funding ratios are in percentage points, AUM in million euro, bond and liability duration in years. The cross-sectional mean, standard deviation, and median are reported. The quarterly sample period is 2009q1-2019q4.

4. Empirical Methodology

I turn to the main tests of the empirical predictions from my theoretical framework next. For bond holdings, I use the *notional* amounts in all my analyses such that I capture active choices by investors and market prices are not driving the results. Moreover, in my main analysis, I focus on maturity buckets to take account of the fact that bonds with similar maturities are close substitutes. I therefore refer to bonds with maturities well above 20 years as “30-year bonds” that are defined as bonds with a remaining time-to-maturity of at least 30 years. I define bonds with maturities close to 20 years as those with remaining times to maturities between 15 and 25 years.

4.1 Long-term bond holdings and the regulatory reform

Panel A of Figure 3 shows the average fraction of P&Is’ total bond portfolio in 30-year bonds. Likewise, panel B shows the fraction of the bond portfolio

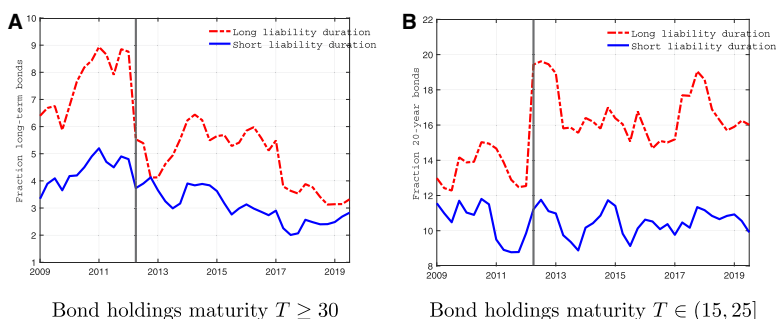


Figure 3
Long-term bond holdings over time

This graph shows the cross-sectional average fraction of the bond portfolio that is invested in 30-year bonds, that is, bonds with a maturity of 30 years or longer (panel A), and for the fraction that is invested in bonds with maturities close to 20 years, that is, between 15 and 25 years (panel B). The sample of P&Is is split in those with long liability durations (above the median) and for those with short liability durations (below the median). The bond portfolio contains the investments in all type of bonds and across all issuer countries. The fractions are in percentage points and the quarterly sample period is 2009q1-2019q4.

with maturities close to 20 years.¹⁹ Both graphs split the sample into P&Is with long (above the median) and short (below the median) liability durations. While both groups show similar trends prior to the reform, the P&Is with long liability durations sharply reduced their allocation to 30-year bonds when the regulatory reform was announced (and implemented). The exact opposite pattern is visible for the allocation toward 20-year bonds, with a sudden and sharp increase in its allocation. Over the sample period, the allocation to 30-year (20-year) bonds remains substantially lower (higher) than the pre-UFR levels.

To align the predictions of the model with the data, I use a difference-in-difference specification which compares long-term bond holdings before and after the implementation of the UFR:

$$w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times \text{UFR}_t + \beta_2 \text{FR}_{it-1}^{-1} + \beta_3 \text{FR}_{it-1}^{-1} \times \text{PF}_i + \beta_4 D_{it-1}^L + \beta_5 \text{AUM}_{it-1} + \beta_6 \text{Vol FR}_{it-1}^{-1} + v_i + \lambda_t + \epsilon_{it}, \quad (6)$$

where w_{it}^B is the bond allocation of P&I i at time t ;²⁰ UFR_t equals one as of the announcement (and implementation) day of the UFR and zero otherwise; $D_{2011q2,i}^L$ is a time-invariant characteristic that equals the liability duration as of 2011q2; FR_{it-1}^{-1} is the lagged inverse of the funding ratio minus one; PF_i is a dummy variable that equals one if the investor is a pension fund; D_{it-1}^L is the lagged liability duration; AUM_{it-1} is the lagged total AUM; $\text{Vol FR}_{it-1}^{-1}$ is the

¹⁹ The bond portfolio contains investments in all type of bonds (e.g., government bonds, corporate bonds) and across all issuer countries.

²⁰ Table IA1 of the Internet Appendix summarizes the dependent variables.

lagged annual volatility of the inverse of the funding ratio; ν_i is the fund fixed effects; and λ_t is the time fixed effects.

I use the liability duration as of 2011q2 in the interaction term to ensure that the liabilities are not in any way affected by the regulatory reform, while at the same time ensuring representativeness of the liabilities as of the regulatory reform. An inverse of the funding ratio equal to one means that P&Is are neither overfunded nor underfunded. Therefore, subtracting one from the inverse of the funding ratio has the advantage that the coefficient is easy to interpret with values above zero indicating underfunding. The regression specification also allows for differences in responses to a decline in funding positions across pension funds and insurers. Although size does not appear in the model, it is added as a control because empirical studies have shown that size is an important driver of investment decisions (e.g., [Pollet and Wilson 2008](#)). Finally, the volatility of the regulatory funding ratio is added as control because the funding ratio alone may not be sufficient to control for P&Is regulatory constraints. The reason is that P&Is that are well-funded but risky may not necessarily be less constrained than P&Is that are less funded but maintain a safer investment strategy.

Table 2 summarizes the results. P&Is with long liability durations decrease 30-year holdings to a larger extent than the ones with short liability durations. At the same time, P&Is increased their bond holdings with maturities varying between 15 and 25 years, while they did not change their holdings of bonds with maturities fewer than 15 years. These results support the first prediction of my theoretical framework in Section 2.

The effects are also economically significant. The total decline in 30-year bond holdings approximately equals $\sum_{i=1}^N \hat{\beta}_1 \times D_{2011q2,i}^L \times AUM_{2012q1,i}^B = \text{€}10.15$ billion, where $AUM_{2012q1,i}^B$ is the total AUM in bonds (nominal terms) for P&I i in the two quarters before the regulatory reform was announced and implemented in 2012q3. The decline is equivalent to a decrease of 33% relative to the pre-reform 30-year bond holdings. Similarly, P&Is increased their 20-year bond holdings by €15.71 billion, or 17% relative to the pre-reform 20-year bond holdings.

The regulatory discount curve is based on the euro swap curve, and hence the suitable interest rates to replicate liabilities are “risk-free” euro interest rates. From a regulatory perspective, the countries that are considered credit-risk free in the regulatory framework are those countries with AAA ratings and that are denominated in euro.²¹ Around the reform, the following euro area countries had AAA ratings and made up the lion share of P&Is’ government bond portfolios: France, Germany, and the Netherlands (Table IA1 of the Internet Appendix).²² Table IA2 of the Internet Appendix replicates the results

²¹ For countries with an AAA rating *not* denominated in euros, a capital charge applies.

²² Austria, Finland, and Luxembourg also had AAA ratings but they did not have debt outstanding with remaining time to maturities of 30 years or longer in the pre-period.

Table 2
Long-term bond holdings and the regulatory discount curve

	Holdings $T \geq 30$		Holdings $15 < T \leq 25$		Holdings $T \leq 15$	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	0.132 [0.35]		-0.832 [-1.11]		1.109 [0.65]	
D_{2011q2}^L	0.375*** [5.24]		0.267 [1.63]		-0.449* [-1.81]	
$UFR \times D_{2011q2}^L$	-0.164*** [-4.54]	-0.157*** [-5.52]	0.153** [2.41]	0.244*** [5.49]	0.148 [1.11]	0.025 [0.28]
D_{t-1}^L	-0.165*** [-2.62]	0.355*** [4.65]	0.047 [0.29]	0.437** [2.05]	0.473** [2.20]	-1.716*** [-7.43]
FR_{t-1}^{-1}	1.683 [0.59]	3.067 [1.20]	-5.253 [-1.05]	-4.995 [-1.20]	-15.092* [-1.87]	-9.946 [-1.48]
$FR_{t-1}^{-1} \times$ Pension funds	-1.018 [-0.31]	2.935 [0.91]	-3.385 [-0.64]	1.428 [0.29]	7.276 [0.80]	-1.218 [-0.16]
log size	0.880*** [3.65]	-1.38 [-1.19]	0.843** [2.29]	-1.992 [-1.64]	-2.871*** [-3.98]	6.207** [2.08]
$Vol\ FR_{t-1}^{-1}$	-0.133 [-1.52]	-0.099 [-1.40]	-0.349** [-2.12]	-0.195* [-1.79]	0.733*** [2.66]	0.281 [1.47]
Life insurance	4.598*** [9.89]		5.427*** [6.56]		-10.149*** [-6.62]	
Pension funds	2.394 [0.79]		5.108 [1.04]		-16.469** [-2.00]	
Fund FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	2,436	2,436	2,436	2,436	2,436	2,436
Adj R^2	.16	.62	.14	.66	.07	.71

This table presents the results of the main regression described in Equation (6): $w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times UFR_t + \dots + \epsilon_{it}$, with the dependent variable equal to the fraction of the P&I's bond portfolio invested in a certain maturity bucket, UFR equal to one as of 2012q3 and zero otherwise, $D_{2011q2,i}^L$ the duration of the liabilities as of 2011q2, and controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. Columns 1 and 2 show the results for bond holdings with a maturity of 30 years or longer; columns 3 and 4 for maturities between 15 and 25 years; and columns 5 and 6 for maturities shorter than 15 years. The bond holdings contain the investments in all type of bonds (e.g., corporate, government) and across all issuer countries. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

for Dutch, French, and German debt only. As expected by focusing on safe euro area government debt, the rebalancing effects are substantially stronger in economic magnitude.

To give additional support for the economic effects, I have also reestimated the regression based on Dutch government bond holdings alone (Table IA3 of the Internet Appendix). The aggregate decline in 30-year Dutch bond holdings equals €2.61 billion. The corresponding amount outstanding equaled €12.13 billion at the implementation of the UFR, and hence, the total decline corresponded to 22% of its amount outstanding. Similarly, the aggregate increase in bond holdings with maturities between 15 and 25 years equaled €4.94 billion, or 26% of its amount outstanding.

My model also predicts that P&Is with long liability durations allocate more to risky assets after implementation of the UFR compared to those with short liability durations. To capture changes in risky asset allocations, I simultaneously study the asset allocation dynamics of stocks, corporate bonds, and government bonds. For corporate and government bonds, I also analyze the evolution of credit risk within both asset classes to distinguish between high risk or illiquid bonds (Ellul et al. 2022). Then, I use the following difference-in-difference specification to formally test the hypothesis:

$$w_{it}^S = \alpha + \beta_1 D_{2011q2,i}^L \times \text{UFR}_t + \beta_2 \text{FR}_{it-1}^{-1} + \beta_3 \text{FR}_{it-1}^{-1} \times \text{PF}_i + \beta_4 D_{it-1}^L + \beta_5 \text{AUM}_{it-1} + \beta_6 \text{Vol FR}_{it-1}^{-1} + v_i + \lambda_t + \epsilon_{it}, \quad (7)$$

where w_{it}^S is the risky asset allocation of P&I i at time t .²³

Table 3 shows the results. The first column confirms that P&Is with long liability durations increase their equity allocation to a larger extent than the ones with short liability durations. A one-standard-deviation increase in the liability duration (6.49) expands the equity allocation by 1.49% but reduces the corporate bond allocation by 1.14%. The allocation to government bonds remains unaltered. Within corporate bonds, there is no evidence of a change in credit risk. On the other hand, a longer liability duration increases the fraction of government bonds allocated to countries with higher credit risk. Column 6 shows that a one-standard-deviation increase in the liability duration increases the fraction of high-yield government bonds held by 1.62%. In sum, these findings show that P&Is move their assets away from corporate bonds to stocks and at the same time increase the riskiness of their government bond portfolios. This finding is consistent with Ge and Weisbach (2021) and Kojien and Yogo (2022), who show that tighter constraints, rather than relaxed ones, result in lower allocations to risky assets.

The final implication for the changes in holdings is that P&Is closer to their solvency constraint decrease (increase) 30-year (20-year) bond holdings to a larger extent than unconstrained P&Is. I use a triple difference-in-difference to test this hypothesis:

$$w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times \text{FR}_{2011q2,i}^{-1} \times \text{UFR}_t + \beta_2 D_{2011q2,i}^L \times \text{UFR}_t + \beta_3 \text{FR}_{2011q2,i}^{-1} \times \text{UFR}_t + \beta_4 \text{FR}_{it-1}^{-1} + \beta_5 \text{FR}_{it-1}^{-1} \times \text{PF}_i + \beta_6 D_{it-1}^L + \beta_7 \text{AUM}_{it-1} + \beta_8 \text{Vol FR}_{it-1}^{-1} + v_i + \lambda_t + \epsilon_{it}, \quad (8)$$

where $\text{FR}_{2011q2,i}^{-1}$ is a time-invariant characteristic that equals the inverse of the funding ratio minus one in 2011q2.

Table 4 has a summary of the results. P&Is that are more constrained, that is, have a larger inverse of their funding ratio, decrease 30-year bond

²³ Table IA1 of the Internet Appendix summarizes the dependent variables.

Table 3
Asset allocation and the regulatory discount curve

	Equity	Corporate bonds		Government bonds		
	Allocation (1)	Allocation (2)	DTD (3)	Allocation (4)	Credit rating (5)	High-yield (6)
$UFR \times D_{2011q2}^L$	0.229** [2.37]	-0.176* [-1.96]	-0.015 [-0.97]	-0.059 [-0.59]	-0.038** [-2.00]	0.250** [2.56]
D_{t-1}^L	0.053 [0.30]	-0.009 [-0.04]	-0.018 [-0.52]	-0.067 [-0.30]	0.032 [0.84]	0.433*** [4.05]
FR_{t-1}^{-1}	4.026 [0.67]	22.099*** [3.14]	1.194 [0.99]	-28.305*** [-4.33]	-0.468 [-0.33]	-7.490*** [-2.70]
$FR_{t-1}^{-1} \times$ Pension funds	-9.835 [-1.40]	-7.038 [-0.90]	-5.347*** [-3.74]	21.204*** [2.66]	-4.589*** [-2.94]	1.323 [0.38]
log size	12.852*** [4.35]	-2.891 [-1.11]	1.272*** [2.77]	-3.356 [-1.03]	2.108*** [3.35]	-5.665 [-1.58]
$Vol\ FR_{t-1}^{-1}$	0.361** [2.17]	0.142 [0.76]	0.031 [0.90]	-0.576*** [-2.87]	0.039 [1.20]	-0.042 [-0.72]
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,436	2,436	2,436	2,436	2,431	2,431
Adj R^2	.83	.77	.65	.72	.59	.37

This table presents the results of the regression described in Equation (7): $w_{it}^S = \alpha + \beta_1 D_{2011q2,i}^L \times UFR_t + \dots + \epsilon_{it}$, with the dependent variable equal to a measure of P&I's risky asset allocation, UFR equal to one as of 2012q3 and zero otherwise, D_{2011q2}^L the duration of the liabilities as of 2011q2, and controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. Column 1 shows the results for the equity allocation, column 2 for the corporate bond allocation, column 3 for the corporate bond distance-to-default measure (DTD), column 4 for the government bond allocation, column 5 for government bond credit risk, and column 6 for the high-yield government bond allocation. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

holdings to a larger extent: a one-standard-deviation increase in the inverse of the funding ratio (0.08), increases the decline in 30-year bond holdings by 1.4% for the P&Is with average liability duration and up to 2.9% for the ones with the longest liability durations, which equals a relative decline of 10%-25%. Furthermore, consistent with the model's predictions, P&Is that are more constrained also increase their holdings of bonds with maturities close to 20 years to a larger extent than unconstrained ones.

The focus in the main specification is on the aggregate bond allocation to specific maturity buckets in order to easily interpret the total magnitude of demand shifts. I show that my results are robust to regressions at the security level in the [Internet Appendix IA5.1.3](#). Moreover, I also show that my results are not driven by QE ([Internet Appendix IA5.1.1](#)) or the fairly long pre and post period used in my main analysis ([Internet Appendix IA5.1.2](#)). Finally, in [Internet Appendix IA5.2](#), I show similar shifts in exposures to long-term interest rates in P&Is swap portfolios.

Table 4
Long-term bond holdings and constraints

	Holdings $T \geq 30$		Holdings $15 < T \leq 25$		Holdings $T \leq 15$	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	-0.491 [-0.40]		-3.226* [-1.82]		0.462 [0.10]	
D_{2011q2}^L	0.509*** [4.54]		0.478** [2.55]		-0.655* [-1.84]	
FR_{2011q2}^{-1}	-16.576 [-1.55]		-51.123*** [-3.01]		43.053 [1.25]	
$D_{2011q2}^L \times FR_{2011q2}^{-1}$	2.335*** [2.59]		2.559** [2.35]		-5.048** [-2.10]	
$UFR \times D_{2011q2}^L$	-0.201* [-1.85]	-0.280*** [-3.92]	0.396*** [3.22]	0.338*** [3.71]	0.154 [0.49]	0.103 [0.56]
$UFR \times FR_{2011q2}^{-1}$	-5 [-0.44]	11.912 [1.37]	-23.211 [-1.21]	5.563 [0.42]	-3.133 [-0.08]	-33.331 [-1.49]
$UFR \times D_{2011q2}^L \times FR_{2011q2}^{-1}$	-0.444 [-0.47]	-1.251** [-2.15]	2.490** [1.99]	1.2528** [1.98]	-0.06 [-0.02]	0.732 [0.53]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
P&I type FE	Yes	No	Yes	No	Yes	No
Fund FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	2,432	2,432	2,432	2,432	2,432	2,432
Adj R^2	.18	.62	.16	.66	.09	.71

This table present the results of the regression described in Equation (8): $w_{it}^B = \alpha + \beta_1 D_{2011q2,i}^L \times FR_{2011q2,i}^{-1} \times UFR_t + \dots + \epsilon_{it}$, with the dependent variable equal to the fraction of the P&I's bond portfolio invested in a certain maturity bucket, UFR equal to one as of 2012q3 and zero otherwise, D_{2011q2}^L the duration of the liabilities as of 2011q2, and FR_{2011q2}^{-1} the inverse of the funding ratio minus 1 as of 2011q2. Controls include the lagged liability duration, the lagged inverse of the funding ratio, the lagged inverse of the funding ratio interacted with a dummy that indicates pension funds, the logarithm of size (AUM), and the annual volatility of the inverse of the funding ratio. P&I type fixed effects include dummies indicating pension funds, life insurers, or nonlife insurers. Columns 1 and 2 show the results for bond holdings with a maturity of 30 years or longer, columns 3 and 4 for maturities between 15 and 25 years, and columns 5 and 6 for maturities shorter than 15 years. The bond holdings contain the investments in all type of bonds (e.g., corporate, government) and across all issuer countries. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered at the investor level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$

5. The Effect of Demand Shifts on Long-Term Yields

In this section, I estimate the effect of the regulatory reform on Dutch government bond yields by using the construction of the UFR as an exogenous shock to demand that affects yields differently at different maturities. I then use this construction as an instrument to estimate price elasticities of demand for various investor types by using the framework of [Kojien and Yogo \(2019\)](#). I also show the implications of the reform for yields in other safe euro area countries and study differences between short-term and long-term price elasticities of demand. Finally, I provide suggestive evidence as to why some investor types are more price elastic in the government bond market than others.

Even though the previous section shows that the demand shift from 30-year to 20-year bonds is visible across various (safe) bonds and not specific to Dutch

debt, here I focus on Dutch government bond yields. The reason is that the demand shock and corresponding price impact were largest for Dutch debt, which allows for a clean identification of price elasticities of demand. This approach implicitly assumes that other investors only absorb the shock in the Dutch government bond market, and not in other safe European bond markets. This is an extreme assumption, because these investors simultaneously operate in the Dutch and other safe European bond markets. In Section 5.3, however, I show that my results do not materially change when I allow investors to operate in both markets.

5.1 Data

To estimate the demand curves, I extend the Dutch holdings data with the ECB holdings data to obtain a larger coverage of Dutch government bond holders. The euro area Securities Holdings Statistics by Sector (SHSS) data provides the sector-country holdings for each sector and country in the euro area over the period 2009q1-2019q4. Like for the Dutch SHS data, per country-sector, the data contains information on the quarter-end holdings at the ISIN level. For instance, the SHSS data reports the aggregate holdings of German banks in a specific security.²⁴ The euro area SHSS data, like the Dutch SHS data, are linked to the CSDB database to obtain market information about the securities held. The five distinct sectors that I incorporate in my analysis are banks, mutual funds, P&I sector (outside the Netherlands), other (households, nonfinancial institutions, etc.), and the foreign sector. The investments of the foreign sector are defined as the difference between the total amount outstanding minus the holdings of all the euro area investors combined.

5.2 The connection between portfolio holdings and yields

As shown in Figure 1, the 30-20 year government bond spread increased significantly after the announcement and implementation of the UFR, and remained at a higher level thereafter. Figure IA4.2 furthermore shows that the observed yield curve moved from being inverted at the longer end to become upward-sloping after the reform.²⁵ These findings are consistent with the model-implied yields as laid out briefly in Section 2 and in detail in Internet Appendix IA4.2, where I close the model by introducing an arbitrageur and impose market clearing.

Next to showing the effect on the yield curve, the goal of this section is to provide insight into the mechanism behind the yield effects. So even though the model gives suggestions about the change in the slope of the yield curve, it does not directly speak to the magnitude of the effects nor the investors that are

²⁴ For more details on the euro area SHSS data, see, for instance, Regulation (EU) No 1011/2012 (<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32012R1011>), and Kojien et al. (2017, 2021).

²⁵ Looking at yield spreads is a cleaner way to identify the effect of the regulatory reform, because movements in yields that are common across maturities cancel out. Nevertheless, both figures show an increase in the slope of the yield curve at the longer end.

driving it. To get more insight in the mechanism, I estimate the demand curves of the other investors that hold Dutch debt and thus have to absorb the demand shock that is caused by the Dutch P&I sector. Using the demand curves, I can measure their price elasticities of demand which, in turn, allow me to study price effects as well as the importance of various investor types in creating price effects.

To estimate demand curves, I apply the asset demand system developed by [Kojen and Yogo \(2019\)](#). Formally, investor i 's investment in Dutch government bonds within maturity bucket h is denoted by $B_{it}(h)$, and the investment in the outside asset is denoted by O_{it} .²⁶ I cannot observe what investors consider to be the outside asset, so I use the 10-year German yield as a proxy for the outside asset because German government bonds are likely good substitutes for Dutch ones.

The portfolio weight in the framework of [Kojen and Yogo \(2019\)](#) is then defined as:

$$w_{it}(h) = \frac{B_{it}(h)}{O_{it} + \sum_{h=1}^{H=7} B_{it}(h)} = \frac{\delta_{it}(h)}{1 + \sum_{h=1}^{H=7} \delta_{it}(h)}, \quad (9)$$

where $\delta_{it}(h) = w_{it}(h)w_{it}^{-1}(0) = B_{it}(h)O_{it}^{-1}$ and $w_{it}(0) = 1 - \sum_{h=1}^{H=7} w_{it}(h)$ equals the fraction invested in the outside asset. The demand of investor i for government bonds with maturity h is a function of bond yields and characteristics ([Kojen et al. 2021](#)):

$$\begin{aligned} \ln B_{it}(h) &= \ln \delta_{it}(h) + \ln O_{it} \\ &= \hat{\alpha}_i + \beta_{0i} y_t(h) + \beta'_{1i} \mathbf{x}_t(h) + \hat{\beta}_{2i} y_t^{DE} + \beta_{3i} \ln(B_{2009q1,i}(h)) \\ &\quad + \epsilon_{it}(h), \end{aligned} \quad (10)$$

in which $\hat{\alpha}_i = \alpha_i + \ln O_i$, $\hat{\beta}_{2i} = \beta_{2i} + \psi_i$, $y_t(h)$ is the average yield for maturity bucket h , $\mathbf{x}_t(h)$ represents bond characteristics, and y_t^{DE} is the 10-year German yield and captures alternative investment opportunities outside of the Netherlands.²⁷ Moreover, the inclusion of the initial holdings, $\ln(B_{2009q1,i}(h))$, captures time-invariant omitted characteristics.

[Kojen and Yogo \(2019\)](#) show that (10) is consistent with a model in which investors have mean-variance preferences over returns, assume that returns follow a factor model, and assume that both the expected returns and factor loadings are affine in a set of characteristics. The component of demand that

²⁶ Unfortunately, the data does not allow me to incorporate holdings of interest rate derivatives to estimate demand curves, because the instrument requires data before and after the reform and the detailed derivatives data only became available in 2017. However, as shown in [Internet Appendix IA5.2](#), the swaps are used in a similar way as bonds: P&Is use swaps to increase exposures toward the 20-year interest rate, which provides suggestive evidence of similar estimates for the demand curves in case one would include derivative positions.

²⁷ The assumption that holdings of the outside asset move only due to changes in the German yield result in $O_{it} = O_i \exp(\psi_i y_t^{DE})$ and hence the natural logarithm results in the terms $\ln O_i$ and $\psi_i y_t^{DE}$ that can be placed on the right-hand side of Equation (13).

is not captured by prices, characteristics, and time-invariant characteristics, $\epsilon_{it}(h)$, is referred to as latent demand. The underlying assumption for Equation (10) to hold is that supply is exogenously given or fixed and that other bond characteristics have not changed because of the reform. I provide evidence that both assumptions hold in my setting in [Internet Appendix IA8](#).

Moving to the bond characteristics, I assume that yields are primarily driven by duration and convexity (measured as the duration squared), because Dutch government bonds are regarded as safe (long-term) assets.²⁸ I also add the average coupons and the total amount outstanding (TAO) for each maturity bucket as characteristics that drive demand for bonds. Investors who aim to match their cash flows might have a preference for coupon bonds and likewise, others may have a preference for more or less liquid bonds proxied by TAO. For the investors, I aggregate the holdings of all investors within a sector as in [Koijen et al. \(2021\)](#).²⁹

To obtain consistent estimates of the parameters in (10) using ordinary least squares (OLS), one has to assume that characteristics are exogenous to latent demand. However, a positive latent demand for Dutch government bonds of a particular maturity may result in lower yields. The demand curves are therefore estimated using an instrumental variable approach. [Koijen and Yogo \(2019\)](#) propose an instrument that is not ideal in government bond markets. Their instrument uses the investment mandates of other investors to construct an instrument that is related to yields but orthogonal to latent demand. The reason is that the investment universe for government bonds is not as restricted and predetermined as for the equity ([Koijen and Yogo 2019](#)) and corporate bond market ([Bretscher et al. 2024](#)). Indeed, [Table IA5](#) of the [Internet Appendix](#) shows that only 53% of the government bonds that are currently held in the portfolios of euro area investors were in the portfolio the previous quarter.

I therefore take advantage of the regulatory reform and use the weights assigned to the UFR as an instrument for changes in yields. I justify the plausible exogenous nature of the reform as follows. Even though investors may have anticipated the UFR, they did not know the determinants of the shape of the UFR, such as its level and the slope. In fact, in discussions between the regulator and the P&I sector about the UFR in the weeks before its implementation, the regulator prohibited P&Is to trade on this information. One specific insurer, Delta Lloyd, did not comply and as a result, received a €22.7 million fine.³⁰ As a result, the demand shift by the P&I sector is close to exogenous and can be used to estimate the demand curves of the other investors, because the demand shift only affects holdings of other investors through its effect on prices. To the extent that P&Is anticipated a lower sensitivity of their liabilities to very

²⁸ Indeed, the Fitch credit rating for the Netherlands is AAA over my entire sample period.

²⁹ [Table IA4](#) reports the summary statistics on bond characteristics.

³⁰ For details, see the [Rotterdam District Court \(2015\)](#).

Table 5
Instrument for every maturity bucket

	(1, 5]	(5, 10]	(10, 15]	(15, 20]	(20, 25]	(25, 30]	(30, ∞)
$\xi(h)$	0	0	0	-0.41	-0.15	0.58	0.91

This table shows the value of the instrument for each maturity bucket used in the instrumental variable approach. The instrument is constructed as the average weight assigned to the UFR for each maturity bucket, minus the average weight assigned to the 20-year interest rate equally distributed over the (15, 20] and (20, 25] maturity buckets. An overview of the weights for each separate maturity is given in Table IA2.

long-term interest rates, the estimated changes in yields may understate the true effects and the demand elasticities that I estimate provide an upper bound of the true elasticities.

The instrument is constructed in such a way that its negative for the maturity buckets (15, 20] and (20, 25] because of the excess demand for 20-year bonds, whereas the instrument is positive for the maturity buckets (25, 30] and (30, ∞) because of the reduced demand for 30-year maturities. Specifically, the instrument is constructed as the average weight assigned to the UFR for each maturity bucket summarized in Table IA12 of the Internet Appendix, minus the total weight assigned to the 20-year interest rate equally distributed over the (15, 20] and (20, 25] maturity buckets. The instrumental variable is then defined as $z_t(h) = \xi(h)UFR_t$, where $\xi(h)$ is the average weight assigned to the UFR for maturity bucket h .

The first-stage regression of the instrumental variable estimator is summarized in panel A of Table 6 for all sectors and equals:

$$y_t(h) = \gamma_{0i} + \gamma_{1i}z_t(h) + \gamma'_{2i}\mathbf{x}_t(h) + \gamma_{3i}y_t^{DE} + \gamma_{4i}\ln(B_{2009q1,i}(h)) + \epsilon_{it}(h). \quad (11)$$

The first-stage coefficient estimates range from 0.28 to 0.33 and the corresponding t -statistic from 5.09 to 7.06, and are therefore higher than the proposed threshold of 4.05 by Stock and Yogo (2005) for rejecting the null of weak instruments at the 5% level. A coefficient of 0.30 means that government bond yields with maturities between 16 and 20 years decreased by 12 basis points and maturities between 21 and 25 years decreased by 5 basis points but maturities between 26 and 30 year went up by 17 basis points and maturities longer than 30 years went up by 27 basis points.

Columns 1-5 of Table 6 show the estimates of the demand system. The demand curves for all investors are downward sloping, except they are upward-sloping for the P&I sector outside the Netherlands. Moreover, all investor types prefer bonds with large outstanding amounts (liquid bonds). The P&I sector prefers bonds with long durations, and the foreign investors and banks prefer bonds with strong convexity. Banks and the foreign sector move away from Dutch government bonds when the German yield goes up, whereas the P&I sector instead moves simultaneously toward Dutch government bonds.

In Internet Appendix Tables IA6 and IA7, I show that my results are robust to using an instrument that uses high frequency changes in long-term yields around the announcement date of the reform as an alternative instrument

Table 6
Demand system: Regulatory reform as instrument

A. First-stage IV					
	Foreign $y_t(m)$	Bank $y_t(m)$	MF $y_t(m)$	P&I excl. NL $y_t(m)$	Other $y_t(m)$
$z_t(m)$	0.2806*** [5.87]	0.3287*** [5.09]	0.2766*** [6.55]	0.2824*** [7.06]	0.2902*** [6.30]
Bond controls	Yes	Yes	Yes	Yes	Yes
Initial holdings	Yes	Yes	Yes	Yes	Yes
Observations	260	213	260	260	245
Adj R^2	.97	.97	.97	.97	.97
Cragg-Donald statistic (F -stat)	20.38	18.00	21.30	23.07	20.37
B. Second-stage IV					
	Foreign holdings	Bank holdings	MF holdings	P&I excl. NL holdings	Other holdings
$y_t(m)$	0.5395* [1.68]	2.3526** [2.05]	0.0184 [0.08]	-0.5533** [-1.98]	0.2419 [0.84]
Duration	-0.3421*** [-3.21]	-0.7946** [-2.34]	0.0179 [0.28]	0.1333** [1.97]	-0.1675 [-0.81]
Convexity	0.0095*** [3.26]	0.0217** [2.37]	-0.0004 [-0.25]	-0.0048** [-2.35]	-0.0031 [-0.52]
Coupon	-0.0950*** [-3.67]	0.0098 [0.15]	-0.1095*** [-5.45]	0.0055 [0.20]	0.0311 [0.76]
AUM outstanding	1.3936*** [20.55]	0.9013*** [3.07]	0.7376*** [11.92]	0.2178*** [4.85]	0.3216** [2.18]
10-year German yield	-0.7984** [-2.28]	-2.9318* [-1.86]	0.1724 [0.65]	0.5812* [1.90]	0.0835 [0.10]
Initial holdings	-0.3312*** [-4.58]	0.6789*** [3.63]	0.2102*** [2.92]	0.4873*** [6.05]	0.2465** [2.29]
Observations	260	213	260	260	245
Adj R^2	.92	.76	.92	.87	.92

This table shows the regression results of the demand system described in Equation (13): $\ln B_{it}(h) = \hat{\alpha}_i + \beta_{0i} y_t(h) + \beta'_{1i} x_t(h) + \hat{\beta}_{2i} y_t^{DE} + \beta_{3i} \ln(B_{2009q1,i}(h)) + \epsilon_{it}(h)$. Panel A shows the first stage of the IV and panel B the second stage. The instrument $z_t(h)$ equals the weights assigned to the UFR for each maturity bucket h interacted with a dummy that equals one after implementation of the UFR. Bond characteristics include the average bond duration, convexity, coupon, and the logarithm of AUM outstanding. The outside asset O_{it} is proxied by the 10-year German yield and initial holdings are added as control to capture time-invariant omitted characteristics. The quarterly sample period is 2009q1-2019q4. The standard errors are clustered by quarter, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

for bond yields. Moreover, Internet Appendix IA7 shows that the ranking of investors based on their price elasticities of demand is similar when using the traditional instrument proposed by Koijen and Yogo (2019).

Next, I can use the demand system to connect prices to the elasticity of demand by taking the derivative of quantities with respect to price for all investor types (Koijen and Yogo 2019; Koijen et al. 2021):

$$-\frac{\partial q_{it}(h)}{\partial p_{it}(h)} = 1 + 100 \frac{\beta_{0i}}{T_i(h)} (1 - w_{it}(h)), \tag{12}$$

Table 7
Price elasticity of demand

	Obs	Median	SD	p5	p95	Weights
Banks euro	213	14.95	12.58	8.61	47.99	15
Foreign investors	260	3.53	2.21	2.11	9.35	55
Mutual funds euro	260	1.11	0.11	1.06	1.41	16
P&I euro (non-NL)	260	-2.07	4.21	-14.22	-0.75	10
Other euro	245	1.63	1.23	1.03	4.78	4
Total		4.31				100

This table shows the price elasticity of demand, computed as in Equation (12): $-\frac{\partial q_{it}(h)}{\partial p_{it}(h)} = 1 + 100 \frac{\beta_{0t}}{T_t(h)} (1 - w_{it}(h))$ for each investor type, maturity bucket h , and quarter t . The median, standard deviation, 5th percentile, and 95th percentile are provided. The total elasticity is the weighted median elasticity, using the weights of each sector defined in the last column. The quarterly sample period is 2009q1-2019q4.

where lowercase means the logs of variables, and $T_t(h)$ is the average maturity for maturity bucket h . To compute $w_{it}(h)$, I take the weight of sector i in Dutch government bonds within maturity bucket h at time t , relative to the total investments in all government bonds at time t .

The demand elasticities with respect to price for each investor type are summarized in Table 7. A demand elasticity close to zero indicates that demand is inelastic, and a large value indicates that demand is sensitive to the price. Banks have the highest demand elasticity, followed by foreign investors. The market clearing condition means that the weighted average elasticity matters for deriving yield effects and equals 4.31. The price elasticity ranges between 4.64 and 5.64 when using an alternative instrument that uses high frequency changes in long-term yields around the announcement date of the reform (Table IA8).

As in [Kojien et al. \(2021\)](#), demand elasticities are substantially higher than the estimates for stock markets; for example, [Chang, Hong, and Liskovich \(2015\)](#) find an elasticity close to one. However, I find that banks are more elastic than mutual funds, while [Kojien et al. \(2021\)](#), who estimate demand elasticities at the issuer-country level, find the opposite. In the next section, I will discuss reasons why banks might be substantially more elastic across maturity buckets of (safe) debt relative to the other investor types.

To derive asset pricing effects from the demand system, I can perform a simple back-of-the-envelope calculation. Pension funds and insurers sold 22% of the amount outstanding of 30-year Dutch government bonds. This percentage means a price effect equal to $22\%/4.31 = 5.11\%$. For a bond with a maturity of 30 years, this percentage means an increase in 30-year yields of 17 basis points, which is close to the price effect found for the first-stage regression.

5.3 Accounting for other safe euro area government bonds

So far, I have shown that the regulatory reform affected the long-end of the Dutch yield curve. However, P&Is did not only rebalance their Dutch

government bond portfolio but also sell 30-year bonds and buy 20-year ones across various fixed income assets. In particular, as shown in [Internet Appendix Table IA1](#), the other two main fixed income assets that P&Is hold are German and French government bonds. [Figure IA1](#) of the [Internet Appendix](#) shows that as a fraction of the amount outstanding, P&Is also reduced their 30-year bond holdings in France and Germany.³¹ Consistent with this selling behavior across various safe euro area countries, [Greenwood and Vissing-Jorgensen \(2018\)](#) show that the 30-10 year yield spreads in other safe euro area countries went up by 8 basis points around the announcement date of the reform, which is half the magnitude of the effect on Dutch debt.

The fact that the price effect was substantially larger for Dutch bonds compared to other safe euro area bonds already suggests that markets are segmented. Otherwise the yields in all safe euro area countries would move in lockstep, as any price divergence between different safe euro area bonds would directly be arbitrated away. In my analysis, I implicitly assume an extreme form of market segmentation, because I only allow the other investors to operate in the Dutch market. This is an extreme assumption, because these investors also operate in other safe euro area bond markets. In addition, the fact that Dutch P&Is also moved from 30-year to 20-year bonds in other safe euro area bond markets implies that the other investors had to absorb the demand shock not only in the Dutch debt market, but also in these other safe euro area bond markets.

To incorporate this idea in the demand system, I separate the Dutch yield $y_t^{NL}(h)$ such that $y_t^{NL}(h) = y_t^{NL,S}(h) + y_t^{EU}(h)$, where $y_t^{NL,S}(h)$ equals the difference between the Dutch yield and the safe euro yield $y_t^{EU}(h)$. Specifically, in the demand system we get:

$$\begin{aligned} \ln B_{it}(h) &= \ln \delta_{it}(h) + \ln O_{it} \\ &= \hat{\alpha}_i + \beta_{0i} y_t^{NL,S}(h) + \beta_{1i} y_t^{EU}(h) + \beta'_{2i} \mathbf{x}_i(h) + \hat{\beta}_{3i} y_t^{US} \\ &\quad + \beta_{4i} \ln(B_{2009q1,i}(h)) + \epsilon_{it}(h). \end{aligned} \quad (13)$$

In words, demand for Dutch bonds now depends on the safe euro yield ($y_t^{EU}(h)$) and the difference between the Dutch and the safe euro yield ($y_t^{NL,S}(h)$). In the data, I compute $y_t^{EU}(h)$ as the value-weighted yield of safe European countries other than the Netherlands. I then compute the Dutch yield spread $y_t^{NL,S}(h)$ as the difference between the Dutch yield and the corresponding European counterpart. For the outside asset, I take the 10-year U.S. government bond yield.³²

³¹ Other safe euro area countries at the time of implementation of the reform are Luxembourg, Finland, and Austria. However, they had limited 30-year debt outstanding in the pre-period, which makes it impossible to compare 30-year bond holdings before and after the reform.

³² The results do not depend on the exact choice of the outside asset, as including time fixed effects gives similar results.

To estimate the demand system, I construct an instrument for both $y_i^{NL,S}(h)$ and $y_i^{EU}(h)$. I cannot use the regulatory reform directly, because it is not country-specific. Instead, I focus on an alternative instrument that uses the high-frequency changes in yields around the event day. Specifically, the instrument for $y_i^{EU}(h)$ equals the value-weighted changes in yields in safe European countries, except for the Netherlands, in a 1-day window around the announcement and implementation day of the regulatory reform ($z^{EU}(h)$ in Table IA6). The instrument for $y_i^{NL,S}(h)$ is the difference in the yield changes observed in the Netherlands, and those changes observed for the value-weighted safe European bond portfolio in the same 1-day window (the difference between $z^{NL}(h)$ and $z^{EU}(h)$ in Table IA6). The instrument satisfies the exclusion restriction under the assumption that the regulatory reform was the only event affecting safe long-term yields around the event day. To be conservative, I therefore focus on 1-day yield changes (rather than lower frequencies).

Notice that conceptually, this approach is very similar to Chaudhary, Fu, and Li (2023), who use a bond's own idiosyncratic mutual fund flow shocks to back out the price multiplier of the bond's own demand shock, while they use the value-weighted idiosyncratic mutual fund flow shocks of a substitute portfolio of other, similar bonds to control for the substitute portfolio. Translating this to my setting, when we observe a demand shock to the Dutch portfolio, it is plausible that other similar bonds may also experience demand shocks, which is consistent with the finding in the first part of the paper that Dutch P&Is sell not only Dutch bonds but also other safe European bonds.

Table A1 summarizes the results. The demand elasticities with respect to the Dutch yield spread are close to the previous estimates of investors' sensitivities to the Dutch yield without controlling for the substitute portfolio. Reestimating the price elasticities of demand in Table A2, I find an aggregate demand elasticity that is equal to 5.56, compared to an aggregate demand elasticity of 5.64 when I do not control for the substitute portfolio.³³ This finding implies that markets are segmented, because if investors would treat Dutch debt as being similar to other safe European debt, then the sensitivity to changes in the Dutch yields spread should be very high (i.e., $\beta_{0i} \rightarrow \infty$ and hence $-\frac{\partial q_{it}(h)}{\partial p_i(h)} \rightarrow \infty$).

However, these results have to be interpreted with care. The reason is that the Cragg-Donald Statistic only passes the threshold of 10 for rejecting weak instruments for the mutual fund sector (Stock and Yogo 2005). The Cragg-Donald Statistic for the other sectors varies between 6.49 and 8.63, which implies that the maximal bias of the IV relative to OLS is 20% or less (rather than 10% or less when meeting the threshold of 10). I therefore leave

³³ Notice that the elasticity of 5.64 comes from the demand estimation in Table IA8 for an accurate comparison using the instrument based on yield changes in a 1-day window.

a deeper analysis of estimating cross-elasticities across various safe European bond markets to future research.

5.4 Short-term versus long-term demand elasticities

So far, the analysis does not distinguish between short-term and long-term price elasticities of demand. However, if investors display inertia in their portfolio allocation decisions as is well-documented for the household sector (e.g., Brunnermeier and Nagel 2008; Andersen et al. 2020; Gabaix et al. 2023), long-term demand elasticities will be higher than short-term demand elasticities because investors react to price changes with a lag.

To study short-term versus long-term price elasticities of demand in the demand system, I rewrite Equation (13) following Gabaix et al. (2023) as:

$$\ln B_{it}(h) = \phi \ln B_{it}^A(h) + (1 - \phi) \ln B_{it}^I(h), \quad (14)$$

where $\ln B_{it}^A(h)$ equals the portfolio allocation when the investor rebalances following the demand system in Equation (13) and $\ln B_{it}^I(h) = \ln B_{it-1}(h)$ the portfolio allocation when the investor does not rebalance.³⁴

In this setting, the long-term demand elasticity equals $\frac{\partial q_{it}(h)}{\partial p_{it}(h)}$ and the short-term demand elasticity equals $\phi \frac{\partial q_{it}(h)}{\partial p_{it}(h)}$. Estimating Equation (14) only requires to control for the lagged portfolio allocation $\ln B_{it-1}(h)$ in Equation (13). The coefficient for the lagged portfolio allocation equals $1 - \phi$, from which we in turn can obtain ϕ to estimate the short-term and long-term demand elasticities.

Table A3 of the appendix summarizes the results. The short-term demand elasticities are close to the elasticities estimated in the main specification (Table 7) with an aggregate short-term price elasticity of demand of 4.87. This result is intuitive, because in the main specification I only estimate the direct response to price changes, that is, the short-term elasticity. The long-term demand elasticity is substantially higher and equals 9.26, or the short-term elasticity divided by the aggregate persistency $\theta = 0.5$ across investors. This implies that in the long run the increase in 30-year yields equals approximately 8.5 basis points, compared to the short run increase of 17 basis points.

There exists substantial heterogeneity in the persistency parameter θ across investors. The estimates reveal that banks are least portfolio inert ($\theta = 0.67$), whereas the P&I sector is most portfolio inert ($\theta = 0.26$). The latter finding is not surprising because of the buy and hold strategies often implemented by this sector.

5.5 Why are banks the most price elastic?

Interestingly, as shown in Figure 3, banks substantially lowered their exposure to 20-year bonds after the regulatory reform while at the same time they

³⁴ Because my analysis is based on *nominal* bond holdings, I do not have to take valuation effects into account (i.e., my setting is free from the concern that holdings shift mechanically because of asset returns). In other words, if the investor does not actively rebalance it means that $B_{it}^I(h) = B_{it-1}(h)$.

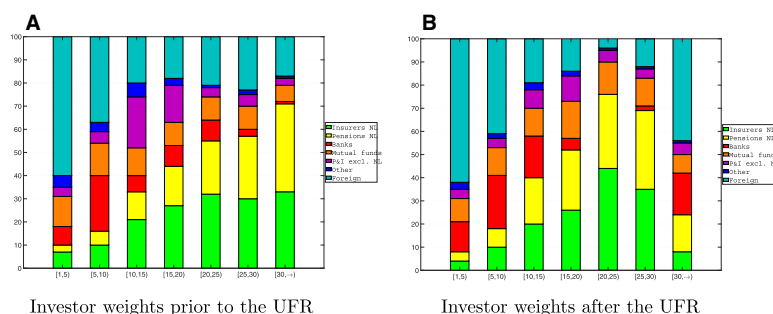


Figure 4
Weights of investor types within maturity buckets

This figure displays the average weights of the investor types (banks, insurance companies, foreign investors, mutual funds, pension funds, and other investors) that held Dutch debt the year prior to the regulatory reform 2011q1-2012q1 (panel A) and the year after the regulatory reform 2012q3-2013q3 (panel B). The banks, mutual funds, and other investor types are at the euro area level. Dutch pension funds, Dutch insurance companies, and the P&I sector in the euro area excluding the Netherlands are separately reported. The foreign sector is determined as the fraction of total amount outstanding that is not held by euro area investors. The weights are in percentage points.

increased their holdings of 30-year bonds, which is consistent with banks being most price elastic. The percentage of TAO held by each investor type is based on the average holdings the year prior and after the reform, however, the figures look similar regardless of the specific quarter, which shows that banks are not just temporary liquidity providers. Quantitatively, banks moved from holding 9% of the total amount outstanding of bonds with maturities close to 20 years to 0% after the regulatory reform. Likewise, they moved from holding 1% of the total amount outstanding of bonds with maturities exceeding 30 years to 18% after the reform.³⁵ This finding provides evidence that banks bought 30-year bonds from the P&I sector, while at the same time they sold 20-year bonds to the P&I sector.

This finding raises the natural question of why banks absorb demand shocks by long-term investors, and not other investor types. Figure 4 shows that over 60% of the government bonds with maturities beyond 10 years are held by the P&I sector. This is a conservative estimate, because part of the foreign sector also consists of P&Is, such as those in the United Kingdom and Switzerland that are not part of my data set. These investors do not respond to prices, to the contrary, their (slightly) upward-sloping demand curves show that they move in the same direction as prices. As theoretically shown by Domanski, Shin, and Sushko (2017), demand for bonds by P&Is goes up when interest rates decline to match the duration of their liabilities.

Mutual funds hold another 11% of the Dutch debt outstanding. As shown in Table 7, their price elasticities of demand are downward sloping, but relatively

³⁵ Notice that these findings cannot be explained by QE, because QE was announced in 2014, while the comparison of the investor weights is based on data from 2011 to 2013.

small in magnitude. This implies that mutual funds are not elastic across maturity buckets. Mutual funds are typically evaluated against benchmarks (e.g., [Pavlova and Sikorskaya 2023](#)) and the duration of those benchmarks are therefore predetermined. As such, mutual funds may experience less room to alter the maturity structure of their bond holdings.

The final major sector in government bond markets is the banking sector. Banks, on the contrary, engage in maturity transformation. Their liabilities are short-term, but they have assets which are long-term. Maturity transformation does not expose them to interest rate risk per se, because their liabilities do not move one to one with interest rates because of banks' market power in deposit markets ([Drechsler, Savov, and Schnabl 2021](#)). As such, a change in the maturity structure of assets does not necessarily affect bank equity negatively. Additionally, a market dominated by the P&I sector may give banks an information advantage, because they are closely connected with the P&I sector through their derivative trades. Using derivative positions of all institutional investors established in the Netherlands that are reported as part of the EMIR regulation since 2017, Table A4 shows that over 85% of the derivative contracts of P&Is are with banks.³⁶ In particular, this pattern is consistent across maturities, and thus, P&Is perform the majority of their interest rate hedging activities with banks.

Furthermore, Figure 5 shows that Dutch banks are also on the opposite side of the swap positions of P&Is. The figure shows the net exposure toward

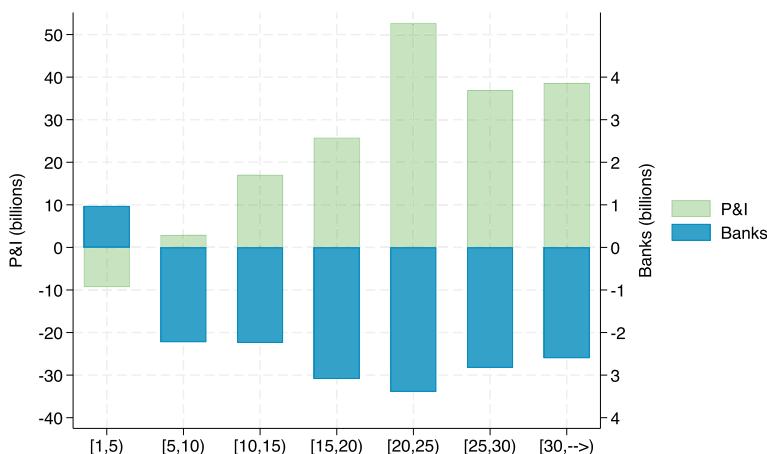


Figure 5
Exposures to interest rates: P&Is versus banks

This figure shows the net exposure of P&Is and banks' swap positions. For the Dutch P&I sector, I compute the aggregate net exposure of their swap positions per maturity bucket. For the Dutch banks, I compute the aggregate net exposures they have with the Dutch P&I sector. The notional amounts to compute the net exposures are based on Euribor plain vanilla swaps. The data come from EMIR and averaged over the period 2018q1-2019q4.

³⁶ [Internet Appendix IA6](#) has a detailed explanation of the data and the cleaning procedure.

different maturity buckets, computed as the total position in receiver swaps minus the total position in payer swaps.³⁷ I then report the results separately for the P&I sector and for the banks' positions with the P&I sector. Interestingly, banks have the exact opposite exposures compared to the P&I sector. In particular, they have the largest negative net exposure to the 20-year interest rate, whereas the P&I sector has the largest positive net exposure to this rate.³⁸ Based on these findings, it is plausible to assume that banks have detailed information about P&Is' demand for specific maturities, which in turn allows them to react quickly when demand shocks occur.

6. Effects of the UFR at the European Level

In this final section, I examine the impact of the UFR implementation at the European level on bond yields in other European countries. Additionally, I analyze bond holdings data for the insurance sector in euro area countries to reveal a significant surge in demand for 20-year bonds, consistent with the patterns observed for Dutch P&Is.

6.1 Yield effects at the European level

The UFR is an important aspect of the EU Solvency II regulation, which harmonized insurance regulation across the EU, that was announced in August 2015 and took effect as of January 1, 2016. Relying on the home-bias found in euro area investors' portfolios (e.g., [Kojen et al. 2017, 2021](#)) and the fact that insurance companies in the Netherlands are subject to similar regulations as other countries in Europe, one may expect similar effects on the yield curve in other European countries, too.³⁹ To test for the effect of the UFR on a broader scale, I construct a panel that comprises 20 European countries subject to Solvency II regulations and regress the yield spreads of these countries on a proxy for insurance demand in those countries that is interacted with a dummy that equals one after the announcement of the UFR.⁴⁰ However, the results that follow should not be interpreted as causal, because by the time Solvency II was announced in August 2015, it was already evident that the UFR would be incorporated into the new regulatory framework. This is different from the Netherlands, which announced the implementation of the UFR 3 years before it

³⁷ Throughout I assume that the floating leg of the swap has zero interest rate risk.

³⁸ Notice that in economic magnitude the total exposures of P&Is are substantially larger than the exposure of Dutch banks to the Dutch P&I sector. The reason is that the P&I sector also performs a substantial amount of their derivative trades with non-Dutch banks as shown in [Table A4](#).

³⁹ The level and the convergence of the regulatory discount curve to the UFR is identical for euro area countries but deviates for countries outside the euro area. In particular, the level of the UFR depends on the inflation level for non-euro-area countries. For more details, see, for example, [EIOPA \(2017, p. 8/135\)](#).

⁴⁰ The UFR dummy equals one as of the announcement of the UFR in August 2015, except for the countries Denmark, the Netherlands, and Sweden, because those countries implemented the UFR earlier, and hence, the UFR dummy equals one as of June 2012, July 2012, and February 2013, respectively.

was officially announced by EIOPA. Hence, anticipation effects are much less of a concern for my main analysis compared to the external validity exercise I perform here.

The following countries are included in the panel over the period from 2006 to 2020: Austria, Belgium, Bulgaria, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Slovakia, Spain, Sweden, and the United Kingdom. The size of the insurance sector differs substantially in these countries (Scharfstein 2018; Greenwood and Vissing-Jorgensen 2018). Therefore, I use the size of the insurance sector in each country relative to its debt as a proxy for insurance demand for long-term bonds. The larger the size of the insurance sector relative to debt, the higher demand for long-term assets in that country. I obtain the sizes of the insurance sectors from EIOPA Insurance Statistics.⁴¹

In the regressions, I control for other variables that determine yield spreads: the 10-2 year government bond spread, the debt-to-GDP ratio, the CDS spread, and the age of the population (measured by the fraction of the elderly relative to the total population). The 10-2 year government bond spread controls for the slope of the term structure (Scholtens and Tol 1999). The debt-to-GDP ratio controls for the fact that countries with more debt, or a higher supply of bonds, likely have higher yields (e.g., Greenwood and Vayanos 2014). CDS spreads control for the effects of differences in default risk across countries on yield spreads. The age of the population controls for countries with older populations that have more government debt supply and larger term spreads (Guibaud, Nosbusch, and Vayanos 2013). In some of the specifications, country fixed effects are also included to control for omitted persistent country characteristics that potentially affect yield spreads, such as differences in countries' financial systems.

I then run the following regression:

$$y_{c,t}(h) - y_{c,t}(s) = \alpha + \beta_0 \text{SIZEIC}_c^{2015} \times \text{UFR}_t + \beta_1 X_{c,t} + \lambda_t + \nu_c + \epsilon_{c,t}, \quad (15)$$

where $y_{c,t}(h) - y_{c,t}(s)$ is the h minus s year government bond spread in country c at time t ; UFR_t equals one as of August 2015 and zero otherwise (except for Denmark, the Netherlands, and Sweden, where the dummy equals one as of June 2012, July 2012, and February 2013, respectively); SIZEIC_c^{2015} is the measure of the demand for long-term bonds by the insurance sector of country c in 2015; $X_{c,t}$ includes country controls: 10-2 year spread, debt-to-GDP, CDS spread, and age; λ_t are time fixed effects; and ν_c country fixed effects.

Table 8 has a summary of the results. Countries with larger demand for long-term bonds by the insurance sector have higher 30-20 year yield spreads but lower 20-10 year yield spreads. An increase in the 30-20 year spread is consistent with the findings for the Dutch P&I sector. A one-standard-deviation

⁴¹ See https://www.eiopa.europa.eu/tools-and-data/insurance-statistics_en.

Table 8
Effects of the UFR at the European level

	Spread 30-20y		Spread 20-10y		Spread 30-10y	
	(1)	(2)	(3)	(4)	(5)	(6)
UFR	0.1278*** [3.34]		0.1879*** [2.60]		0.3157*** [3.56]	
Size IC market	-0.0596*** [-2.62]		0.0744 [1.50]		0.0148 [0.26]	
Size IC market × UFR	0.0511* [1.81]	0.0656*** [2.63]	-0.1043* [-1.71]	-0.1052** [-2.27]	-0.0532 [-0.75]	-0.0396 [-0.71]
10-2y spread	0.0477*** [7.06]	0.0508*** [9.36]	-0.012 [-0.66]	-0.0216 [-0.94]	0.0357 [1.54]	0.0291 [1.05]
Debt to GDP	0.0007 [1.55]	0.0017 [1.60]	0.0031*** [4.00]	0.0027 [1.22]	0.0038*** [3.61]	0.0045 [1.60]
CDS	-0.0005*** [-4.23]	-0.0004*** [-4.09]	-0.0009*** [-3.38]	-0.0009** [-2.57]	-0.0013*** [-3.79]	-0.0013*** [-2.93]
Age	-0.0019 [-0.37]	-0.0336 [-1.50]	0.011 [1.09]	0.0234 [0.82]	0.0091 [0.84]	-0.0102 [-0.32]
Country FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	286	286	286	286	286	286
Adj R^2	.83	.87	.42	.53	.71	.75

This table shows the effects of the UFR on yield spreads for a panel of European countries over the period 2006-2020 (annual). The UFR equals one as of the announcement of the UFR as part of the Solvency II regulation in August 2015, except for the countries Denmark, the Netherlands, and Sweden where the dummy equals one as of June 2012, July 2012, and February 2013, respectively. Size IC market equals the size of the insurance market relative to debt in 2015. Controls include the 10-2y government bond spread, debt-to-GDP ratio, CDS spread, and age of the population. Standard errors are clustered at the country level, and the corresponding t -statistics are in brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

increase in insurance demand (0.715), increases the 30-20 year bond spread after the introduction of the UFR by 4.7 basis points and lowers the 20-10 year spread by 7.5 basis points. Countries with a small insurance sector such as Hungary and Portugal experienced negligible changes in yields, while Ireland and Denmark with large insurance sectors experienced a drop of 13-32 basis points in the 20-10 year spread and an increase of 8-20 basis points in the 30-20 year spread, respectively. Table IA9 of the Internet Appendix shows that similar effects occur when I use an alternative measure of insurance demand that multiplies the size of the insurance sector with the aggregated liability duration of insurance companies.⁴²

6.2 Changes in bond holdings by euro area insurance companies

The simultaneous widening of the 30-20 year spread and the reduction in the 20-10 year yield spread, while the 30-10 year spread remains unchanged, suggests that the yield effects are predominantly driven by an increased demand

⁴² Multiplying the size of the insurance sector with the liability duration controls for differences in the liability structure of insurance companies across countries and thereby more accurately measures the demand for long-term bonds.

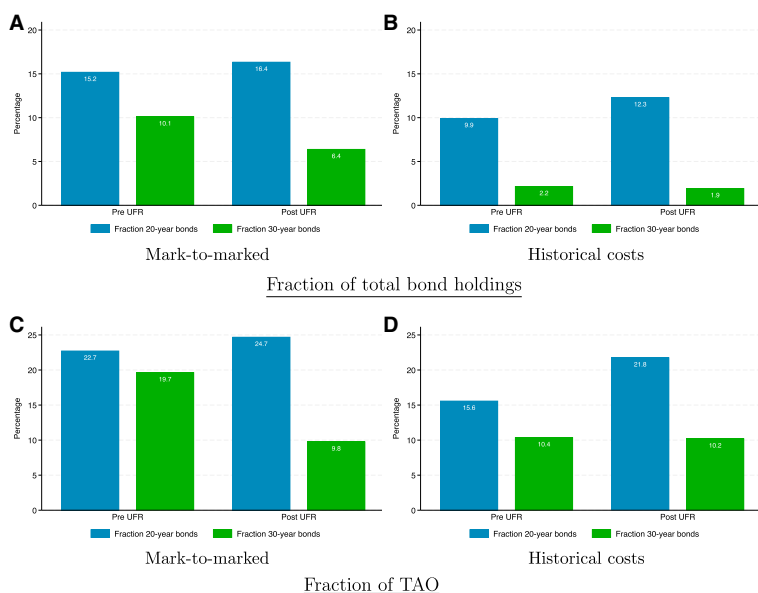
for 20-year bonds. To verify this hypothesis, I turn to the euro area SHSS data used in the previous section which provides information on the bond holdings of the insurance sector in each euro area country.

Before considering the portfolio adjustments, I split countries in two groups. This step is crucial because, in the previous analyses, I implicitly assumed that all countries started from a similar position before the implementation of Solvency II. However, Solvency II introduced a standardized framework for insurance companies across Europe, replacing the previous national regulations. One notable difference was the substantial variation in discounting standards across insurance companies, with many euro area countries relying on historical cost valuation of their liabilities rather than mark-to-market valuation as applied in the Netherlands. As a consequence, the regulatory value of liabilities was not sensitive to market interest rates in those countries that applied historical costs. The theoretical model outlined earlier would therefore suggest that insurance companies' demand for 30-year bonds was likely low to begin with. I therefore separate the countries into those that applied historical costs and those that applied mark-to-market valuation of the liabilities.

Turning to the bond portfolio allocations, Figure 6 shows the allocation to 30-year bonds and those with maturities close to 20 years before and after the reform for those countries that applied mark-to-market valuation (panel A) versus those that applied historical costs (panel B).⁴³ An interesting pattern emerges from this figure. Initially, countries applying historical costs had significantly lower demand for both 30-year bonds and those with maturities close to 20 years. Before the reform, the average allocation to 30-year (20-year) bonds in their portfolios equaled 2% (10%), compared to 10% (15%) for those using mark-to-market valuation. This finding aligns with the notion that the regulatory hedge demand was largely insensitive to market interest rate changes.

Following the reform, countries employing mark-to-market valuation increased their 20-year bond holdings while substantially reducing their 30-year bond holdings. On the other hand, countries relying on historical costs only increased their 20-year bond holdings, with no change observed in their 30-year bond holdings. This observation reinforces the model's prediction: for countries using historical costs, liabilities remained insensitive to 30-year interest rate changes under the new framework, while their sensitivity to 20-year rates increased. Security-level regressions in Table IA10 of the Internet Appendix support these findings, confirming that demand for 30-year bonds remained unchanged while demand for those with maturities close to 20 years increased.

⁴³ Countries that used mark-to-market valuation of the liabilities: Ireland, Luxembourg, and the Netherlands. Countries that used historical costs to value liabilities: Austria, Belgium, Finland, France, Germany, Greece, Italy, Portugal, Slovakia, and Spain. Internet Appendix IA10 provides details about the country classifications.

**Figure 6****Bond holdings pre/post-implementation of the European-wide UFR**

Panels A and B show the average fraction of the bond portfolio invested in 30-year bonds ($T \geq 30$) and the fraction invested in bonds with maturities close to 20 years, that is, $T \in (15, 25]$. The cross-sectional averages are computed both pre- and post-announcement of the European-wide UFR. Panels C and D show the cross-sectional average fraction of the country's bonds outstanding held by the insurance sector in that country, again for the two different maturity buckets. Panels A and C show the results for those countries that applied mark-to-market valuation of the liabilities prior to the UFR (i.e., Ireland, Luxembourg, and the Netherlands) and panels B and D for those that used historical costs valuation (Austria, Belgium, Finland, France, Germany, Greece, Italy, Portugal, Slovakia, and Spain). Finland, Portugal, and Slovakia are excluded from panel D, because for substantial periods of time they are lacking outstanding bonds within either the $T \in (15, 25]$ or $T \geq 30$ maturity bucket. The average allocations pre-UFR (post-UFR) for each country are computed as the time-series average allocation over the period 2009q1-2015q2 (2015q3-2019q4).

To better understand the implications of the findings on yield effects, we must consider the changes in demand relative to the total amount outstanding (TAO) of debt in each country. Panels C and D of Figure 6 illustrate the shift in the fraction of TAO held by insurers in their respective country. That is, I calculate the total bond holdings of the insurance sector in its own country's debt and express it as a percentage of the TAO of debt in that country. Interestingly, a similar pattern emerges: countries that employed mark-to-market valuation of their liabilities held a slightly higher share of the outstanding 20-year bonds, but substantially less of the 30-year bonds. On the other hand, for those that used historical costs, the fraction of outstanding 20-year bonds held by insurers increased, while their holdings of 30-year bonds remained unchanged. This further corroborates the model's predictions and strengthens the link between the observed changes in demand for specific bond maturities and the resultant yield effects found earlier in this section.

In [Internet Appendix IA9](#), I study further implications of the UFR on European yield spreads and asset allocations. In particular, I study the effect of transitionals on yield spreads, where transitionals exempted some countries from using the UFR in the short-term. Moreover, I study the change in risky asset exposures across euro area countries as a result of the implementation of the UFR.

7. Conclusion

In this study, I use holdings data and price data simultaneously to study demand shifts and their causal effect on yields. In particular, I exploit a change in the regulatory discount curve at which the liabilities of long-term investors are evaluated and find a structural change in demand for long-dated assets that led to a downward pressure on 20-year yields but to an upward pressure on longer maturity yields.

Exploiting the heterogeneity in demand shifts across long-term investors shows that constrained investors reacted more heavily to the regulatory reform compared to unconstrained ones which has important implications for the vulnerability of the pension and insurance sector going forward.

My results also show that regulation plays a nontrivial role in the demand for long-term bonds which in turn, affects the yields of these bonds. This finding has direct implications for the role of regulation in determining the government's cost of borrowing.

Finally, by estimating the demand elasticities of investors in the government bond market, I show that the banking sector is most price elastic and primarily responsible for absorbing demand shocks by the P&I sector. My findings therefore suggest that regulators should take the spillover effects to other sectors into account when analyzing policies targeted at long-term investors.

Code Availability: The replication code is available in the Harvard Dataverse at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/YP1D9D>.

Appendix. Additional tables

Table A.1
Demand system: Controlling for a substitute portfolio

	Foreign	Bank	MF	P&I excl. NL	Other
$y_t^{NL,S(m)}$	0.4523* [1.73]	3.1935*** [2.96]	0.2988 [1.02]	-0.7502*** [-2.72]	0.5885* [1.89]
$y_t^{EU(m)}$	-0.8873 [-1.49]	2.5673 [1.47]	0.4069 [0.93]	-0.5976* [-1.73]	-0.124 [-0.16]
Bond controls	Yes	Yes	Yes	Yes	Yes
Initial holdings	Yes	Yes	Yes	Yes	Yes
Observations	258	211	258	258	243
Adj. R-squared	.80	.51	.77	.86	.84
Cragg-Donald statistic (first-stage)	8.63	6.87	10.51	6.49	8.38

This table shows the second-stage estimates of the demand system, allowing for the substitute portfolio to affect investors' holdings. The instrument for the Dutch yield spread ($y_t^{NL,S(m)}$) is $z_{0t}(h)$ and equals the difference between the 1-day yield change in Dutch versus other safe European yields around the implementation date of the regulatory reform. The instrument for safe European bond yields ($y_t^{EU(m)}$) is $z_{1t}(h)$ and equals the 1-day change in yields of the value-weighted European bond portfolio around the regulatory reform. Bond characteristics include the value-weighted average bond duration, convexity, coupon, and the logarithm of AUM outstanding for each maturity bucket. The outside asset O_{it} is proxied by the 10-year U.S. yield and initial holdings are added as control to capture time-invariant omitted characteristics. The quarterly sample period is 2009q1-2019q4. Standard errors are clustered by quarter, and the corresponding t -statistics are in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2
Price elasticity of demand: Controlling for a substitute portfolio

	Obs	Median	SD	p5	p95	Weights
Banks euro	213	23.66	17.87	10.99	65.20	15
Noneuro investors	260	3.12	1.85	1.93	8.00	55
Mutual funds euro	260	2.81	1.87	1.94	7.73	16
P&I euro (except NL)	260	-3.16	5.71	-19.63	-1.38	10
Other euro	245	4.06	3.37	2.40	12.73	4
Total		5.56				100

This table shows the price elasticity of demand, computed as in Equation (12): $-\frac{\partial q_{it}(h)}{\partial p_{it}(h)} = 1 + 100 \frac{\beta_{0t}}{\Gamma_t(h)} (1 - w_{it}(h))$ for each investor type, maturity bucket h , and quarter t , when simultaneously controlling for the substitute portfolio. The median, standard deviation, 5th percentile, and 95th percentile are provided. The total elasticity is the weighted median elasticity, using the weights of each sector defined in the last column. The quarterly sample period is 2009q1-2019q4.

Table A.3
Price elasticity of demand: Short term versus long term

	Obs.	θ	Short-term	Long-term	Weights
Banks euro	197	0.67	11.96	17.79	15
Foreign investors	253	0.46	4.65	10.02	55
Mutual funds euro	254	0.54	2.02	3.75	16
P&I euro (non-NL)	254	0.26	0.64	2.50	10
Other euro	239	0.53	1.31	2.48	4
Total		0.50	4.87	9.26	100

This table shows the persistency in portfolio holdings θ , the corresponding short-term price elasticity of demand $(-\theta \frac{\partial q_{it}(h)}{\partial p_{it}(h)})$ and the long-term price elasticity of demand $(-\frac{\partial q_{it}(h)}{\partial p_{it}(h)})$ for each holder sector i , maturity bucket h , and quarter t , based on estimates that model the partial adjustment process in Section 5.5.4. The medians are reported. Total persistence, short-term demand elasticity, and long-term demand elasticity are the weighted median values, using the weights of each sector defined in the last column. The sample period is 2009q1–2019q4.

Table A.4
Separation of P&I swap portfolio by counterparty

	All	(1, 5]	(5, 10]	(10, 15]	(15, 20]	(20, 25]	(25, 30]	(30, ∞)
Banks euro	34.96	28.67	39.09	35.07	37.01	33.68	40.08	36.94
Mutual funds euro	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P&I euro (except NL)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Pension funds NL	8.47	6.02	9.65	9.18	6.56	10.92	12.15	5.25
Insurance companies NL	3.80	3.32	2.33	4.11	4.69	3.29	6.61	5.52
Other euro	0.54	0.26	0.47	0.49	1.40	0.56	0.94	4.18
Foreign investors:								
Banks	52.23	61.73	48.45	51.14	50.34	51.56	40.23	48.10
Nonbanks	51.42	61.01	47.91	50.83	49.76	50.54	37.96	46.77
	0.81	0.72	0.54	0.31	0.59	1.02	2.27	1.32

This table divides the total notional amount of P&Is' swap positions by counterparty types in percentage points. The sectors are banks in the euro area, mutual funds in the euro area, P&Is in the euro area except those in the Netherlands, pension funds in the Netherlands, insurance companies in the Netherlands, other euro area investors, and foreign investors. Foreign investors are all investors that reside outside of the euro area. The last two rows break up the foreign investors into banks and nonbanks. The results are reported for all swaps and broken down by maturity buckets. The notional amounts are based on Euribor plain vanilla swaps. The data come from EMIR and have been averaged over the period 2018q1–2019q4.

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